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SURVEY

The 2004 IDC Professional Developer Model

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IDC OPINION

IDC has updated its model engine that estimates the number of professional developers worldwide with new benchmark data on software spending, worldwide population, and the global level of tertiary education. IDC now estimates that there were just under 8.7 million professional developers worldwide in 2002 (adjusted up from an earlier estimate of 8.5 million), and that that number increased to just over 9.3 million in 2003, an increase of 7% over 2002. In addition, given the more robust IDC estimate of absolute spending on software for the 2004–2008 forecast period and new information about tertiary education rates, IDC now believes the growth rate in the number of professional developers will increase over time. In 2003, the model engine forecasted that the growth rate — but not the absolute number — would decrease over time. These changes are minimal and some of the differences in forecasts are simply caused by the fact that a different time period is involved. For example, IDC currently forecasts that there will be 13.5 million professional developers worldwide in 2007 (whereas the most recent previous IDC estimate using this model engine forecast 12.9 million professional developers by 2007). The IDC model now forecasts the total will increase to 14.9 million in 2008 but much of that is driven by estimates of worldwide population increases that may be overstated. Details include:

- Regionally, the IDC Professional Developer Model continues to predict that Asia/Pacific will account for approximately 30% of worldwide professional developer head count by the 2008 time frame, up from 21.4% in 2001.
- Software spending per developer continues to trend down over the forecast period on a worldwide basis but it trends up in Western Europe and North America where both IT producers and users will invest more in all aspects of developer automation technology.
- As promised in 2003, we are continuing to watch the correlation of developer headcount growth rate to overall software spending growth rate; another year as proof point indicates that developer headcount growth might be as meaningful to software spending forecasts as a leading indicator as "tool and die" economic activity traditionally was for the manufacturing sector.

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TABLE OF CONTENTS

P

In This Study **1**

Definition 1
Methodology 2
Introduction 2
This Year's Model Compared with All Previous Years' Models 2
This "2004 Model" Compared with the "2003 Model" 2
Countries and Regions Included in This Study 3
Variables in the IDC Professional Developer Model 13
Independent Variables 14
Country Population 14
Tertiary School Education 14
Software Spending 14
Dependent Variables 14
Population-to-Developer Ratio 15
Developer-to-Education Relationship 15
Number of Professional Developers 15
Model Construction 17
The Conceptual Model 17
The Physical Model 20
Forecasting from the Model 23
Summary of Model Assumptions and/or Constraints 24

Situation Overview **25**

Worldwide Professional Developers by Geographic Region 25

Future Outlook **28**

Outlook by Region 44
North America 44
Western Europe and Latin America 44
Asia/Pacific 44
Eastern Europe 44
Middle East and Africa 45
Professional Developer Penetration Ratio by Region 45
Software Spending, Professional Developers, and Productivity 47

Essential Guidance **49**

Learn More **50**

Related Research 50

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LIST OF TABLES

P

1 Worldwide Professional Developers by Geographic Region, 2001–2008 4
2 Worldwide Geographic Regions, Country Affiliations, and Population, 2003 6
3 Dependent Variable: Population to Yield a Developer 15
4 Dependent Variable: Developer-to-Education Relationship 16
5 Dependent Variable: Number of Professional Developers 16
6 Comparison of Actual and Predicted Data for the Number of Professional Developers 21
7 Worldwide Population and Professional Developer Estimates, 2001–2008 26
8 Worldwide Growth in Professional Developers by Geographic Region, 2001–2008 27
9 Worldwide Professional Developers by Country and Region, 2001–2008 29
10 Worldwide Population per Developer by Geographic Region, 2001–2008 46
11 Worldwide Software Spending per Developer by Geographic Region, 2001–2008 48

LIST OF FIGURES

P

1 Modeled Relationship Between Professional Developers and Tertiary Education 18

2 Modeled Relationship Between Professional Developers and Tertiary Education with Fitted Curve 19

3 Known Professional Developers/Education Relationship 22

4 Modeled Professional Developers/Education Relationship 23

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IN THIS STUDY

The IDC Professional Developer Model is a detailed model of the number of professional developers active worldwide beginning in 2001 and extending through 2008. This study answers such questions as: How many professional developers are there worldwide? How is the number of professional developers distributed by geographic region and major country? In what countries and regions will there be strong growth in professional developers? How does software spending correlate to the number of developers?

Definition

IDC defines a developer as an employed or self-employed professional trained or self-taught to use application development and deployment tools as defined in the IDC software taxonomy and related tools available with many packaged applications to build or customize applications used in business, government, nonprofit corporations, and so forth. This includes employees of IT product and service suppliers, as well as employees of their customers and clients. This measurement does not include line managers and proprietors, typically in small businesses or local governments, who "program" systems as an adjunct to their primary position. Included in this definition are individuals in the following professions:

- Computer programmers and system analysts
- Computer software engineers
- Database administrators
- Statisticians and actuaries
- Computer support specialists
- Other computer scientists
- Numerical tool and process control programmers

The definition is based on professions included in U.S. Department of Labor 2000 Standard Occupational Classification (SOC) codes 15-XXXX (**www.stats.bls.gov**). Note that the Department of Labor has changed its classification system from a Standard Industry Classification (SIC)–based taxonomy to these SOC codes since the IDC Professional Developer model was first introduced in 1999, and the Department has changed some occupational-classification names. However the underlying definitions are basically the same for all versions of the IDC model. The U.S. Department of Labor plans to amend or change the system again in 2005.

Methodology

Introduction

The work that IDC has done in determining and segmenting the number of professional developers worldwide is best defined as a model engine in contrast with other forms of sizing the number of developers (e.g., using a census). Although IDC's model does include some census data, it is defined as a model engine because of its reliance on inferred relationships built on top of census data, third-party data collection of other demographic factors such as tertiary education rates, IDC supply-side market data such as estimated and forecasted software spending, IDC empirical data, and so forth.

Although IDC has reviewed the results of this model carefully and is satisfied with our findings, the reader should realize that due to the modeled nature of much of the data, inaccuracies are bound to exist.

This Year's Model Compared with All Previous Years' Models

The IDC Professional Developer Model was first published in December 1999. A heavily revised version of the IDC Professional Developer Model was published in June 2001, with follow-on versions published in December 2002 and December 2003. The 2001 version, while reporting on the same subject (i.e., number of professional developers worldwide by region and country) as the 1999 model, used a far more sophisticated statistical approach. This fifth version of the IDC Professional Developer Model is like the 2001, 2002, and 2003 versions in complexity, and it reflects a number of small yet significant differences introduced in 2002 to provide a higher level of accuracy in the historical and forecast data.

This "2004 Model" Compared with the "2003 Model"

The changes introduced in the 2001 model engine and reflected in this the 2004 version resulted from several attributes, including the following:

- The availability of a wider variety of benchmark data (some in time-series form) regarding the number of professional developers in various countries, including the United States, Canada, the United Kingdom, Japan, India, Germany, Italy, China, Russia, France, Israel, Greece, and Ireland. This data is used to both validate what the model forecasts for those countries and to estimate the number of developers in countries that provide no such statistics. The 2004 model uses this same benchmark data, based on various years' census data from those countries, normalized to 1998 (the most readily available year worldwide, but please note that there are no estimates provided in this document for years prior to 2001, and estimates for those years in previous IDC deliverables should be disregarded because of the changes in historical data).

- Reliance on software spending data and tertiary education to forecast how the number of professional developers changes over time. The software spending data is sourced from IDC's *Worldwide Black Book,* and the tertiary education data is sourced from UNESCO; the 2004 model uses updated data from both

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sources. Note well that two factors have driven up the estimated number of professional developers worldwide as opposed to the 2003 model:

- The IDC forecast of aggregate software spending for the 2004–2008 period is more robust than IDC's forecast of aggregate software spending for the 2003–2007 period because of both the different time period and indications of a healthier rebound from the 2001–2002 IT industry deflation than previously expected, especially in the Asia/Pacific region.

- The UNESCO-based tertiary education rates used in the model engine are substantially different than those used in the 2003 model; new data says that estimated penetration rates are lower in many established economies but higher — from a very low base — in the developing world.

Although the model engine remained almost exactly the same in 2004 as in 2001, 2002, and 2003, the careful reader will note several differences from the results released in December 2003 as a result of the changes to the statistical base (see Table 1):

- The increase in the number of professional developers estimated to be working as of December 2003 (and to have been working in 2001 and 2002)

- An increase in the number of professional developers forecast to be working worldwide in the period from 2004 to 2007

- The addition of a forecast for 2008

Changes in total population up and down also had a minor effect on the forecast changes as they cascade through the model engine.

Countries and Regions Included in This Study

This study provides information about the number of professional developers in most of the over 190 countries that are formally recognized by the United Nations. Of these countries, a subset of countries with large economies has been identified as the baseline from which extrapolations are made to estimate the number of professional developers for the remaining countries. The baseline countries include all the so-called developed countries in the world and account for over 70% of the world's population and over 95% of the worldwide software spending in 2003.

The countries included have been segmented into six distinct geographic regions that align with IDC geographic research configurations. These regions include Asia/Pacific, Eastern Europe, Latin America, the Middle East and Africa, North America, and Western Europe. These regions are mutually exclusive and collectively exhaustive with respect to the countries covered in this study (see Table 2).

TABLE 1

Worldwide Professional Developers by Geographic Region, 2001–2008 (000)

	2001	2002	2003	2004	2005	2006	2007	2008	2001 Share (%)	2003–2008 CAGR (%)	2008 Share (%)
Asia/Pacific											
Primary AP	1,651	1,876	2,058	2,306	2,646	3,052	3,534	4,096	20.1	14.8	27.5
Rest of AP	106	116	121	132	148	174	211	258	1.3	16.3	1.7
Subtotal	1,757	1,992	2,180	2,439	2,794	3,226	3,745	4,353	21.4	14.8	29.2
Eastern Europe											
Primary EE	587	735	911	1,025	1,138	1,254	1,381	1,518	7.2	10.8	10.2
Rest of EE	169	175	181	200	221	240	261	284	2.1	9.4	1.9
Subtotal	757	910	1,091	1,225	1,359	1,494	1,642	1,801	9.2	10.5	12.1
Latin America											
Primary LA	350	331	355	391	435	480	520	560	4.3	9.5	3.8
Rest of LA	230	241	254	277	307	339	369	401	2.8	9.6	2.7
Subtotal	580	572	609	668	743	819	889	961	7.1	9.6	6.5
Middle East/Africa											
Primary MEA	166	174	215	247	283	318	352	384	2.0	12.3	2.6
Rest of MEA	684	734	799	954	1,157	1,379	1,651	1,969	8.3	19.8	13.2
Subtotal	850	908	1,014	1,201	1,440	1,697	2,004	2,353	10.4	18.3	15.8
North America											
Primary NA	2,651	2,635	2,722	2,808	2,904	3,012	3,122	3,226	32.3	3.4	21.7
Subtotal	2,651	2,635	2,722	2,808	2,904	3,012	3,122	3,226	32.3	3.4	21.7

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TABLE 1

Worldwide Professional Developers by Geographic Region, 2001–2008 (000)

	2001	2002	2003	2004	2005	2006	2007	2008	2001 Share (%)	2003–2008 CAGR (%)	2008 Share (%)
Western Europe											
Primary WE	1,606	1,680	1,727	1,805	1,897	1,995	2,096	2,195	19.6	4.9	14.7
Subtotal	1,606	1,680	1,727	1,805	1,897	1,995	2,096	2,195	19.6	4.9	14.7
Worldwide											
Baseline countries	7,012	7,431	7,989	8,583	9,303	10,112	11,007	11,979	85.5	8.4	80.5
Remaining countries	1,189	1,266	1,355	1,564	1,834	2,132	2,492	2,911	14.5	16.5	19.5
Total	8,201	8,697	9,344	10,147	11,137	12,243	13,499	14,890	100.0	9.8	100.0
Growth (%)	NA	6.0	7.4	8.6	9.8	9.9	10.3	10.3			
2002 estimate of professional developers	7,828	8,480	9,332	10,491	11,828	13,315	NA	NA			
2002 model to 2004 model difference (%)	95.5	97.5	99.9	103.4	106.2	108.8	NA	NA			
2003 estimate of professional developers	7,932	8,505	9,041	9,801	10,722	11,751	12,856	NA			
2003 model to 2004 model difference (%)	96.7	97.8	96.8	96.6	96.3	96.0	95.2	NA			

Source: IDC Worldwide Professional Developer Model, 2002, 2003, and 2004

TABLE 2

Worldwide Geographic Regions, Country Affiliations, and Population, 2003

Region	Subregion	Country	Population	Share of Worldwide (%)
Asia/Pacific	Primary AP	Australia*	19,731,984	0.3
Asia/Pacific	Primary AP	China*	1,291,496,022	20.5
Asia/Pacific	Primary AP	Hong Kong*	6,809,738	0.1
Asia/Pacific	Primary AP	India*	1,049,700,118	16.7
Asia/Pacific	Primary AP	Indonesia*	234,893,453	3.7
Asia/Pacific	Primary AP	Japan*	127,214,499	2.0
Asia/Pacific	Primary AP	Korea (South)*	22,466,481	0.4
Asia/Pacific	Primary AP	Malaysia*	23,092,940	0.4
Asia/Pacific	Primary AP	New Zealand*	3,951,307	0.1
Asia/Pacific	Primary AP	Philippines*	84,619,974	1.3
Asia/Pacific	Primary AP	Singapore*	4,276,788	0.1
Asia/Pacific	Primary AP	Taiwan*	22,603,000	0.4
Asia/Pacific	Primary AP	Thailand*	64,265,276	1.0
Asia/Pacific	Primary AP	Vietnam*	81,790,803	1.3
Asia/Pacific	Rest of AP	Bangladesh	138,448,210	2.2
Asia/Pacific	Rest of AP	Bhutan	2,139,549	0.0
Asia/Pacific	Rest of AP	Brunei	358,098	0.0
Asia/Pacific	Rest of AP	Cambodia	13,124,764	0.2
Asia/Pacific	Rest of AP	Fiji	868,531	0.0
Asia/Pacific	Rest of AP	Kiribati	98,549	0.0
Asia/Pacific	Rest of AP	Korea, Dem. Rep. (North)	48,035,733	0.8
Asia/Pacific	Rest of AP	Laos	5,921,545	0.1
Asia/Pacific	Rest of AP	Maldives	329,684	0.0
Asia/Pacific	Rest of AP	Marshall Islands	56,429	0.0
Asia/Pacific	Rest of AP	Micronesia	108,143	0.0
Asia/Pacific	Rest of AP	Mongolia	2,712,315	0.0
Asia/Pacific	Rest of AP	Myanmar	42,510,537	0.7

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TABLE 2

Worldwide Geographic Regions, Country Affiliations, and Population, 2003

Region	Subregion	Country	Population	Share of Worldwide (%)
Asia/Pacific	Rest of AP	Nepal	26,469,569	0.4
Asia/Pacific	Rest of AP	Pakistan	156,127,453	2.5
Asia/Pacific	Rest of AP	Papua New Guinea	5,295,816	0.1
Asia/Pacific	Rest of AP	Samoa	178,173	0.0
Asia/Pacific	Rest of AP	Solomon Islands	509,190	0.0
Asia/Pacific	Rest of AP	Sri Lanka	19,742,439	0.3
Asia/Pacific	Rest of AP	Vanuatu	199,414	0.0
Eastern Europe	Primary EE	Bulgaria*	7,588,399	0.1
Eastern Europe	Primary EE	Croatia*	4,497,779	0.1
Eastern Europe	Primary EE	Czech Republic*	10,251,087	0.2
Eastern Europe	Primary EE	Hungary*	10,057,745	0.2
Eastern Europe	Primary EE	Poland*	38,622,660	0.6
Eastern Europe	Primary EE	Romania*	22,380,273	0.4
Eastern Europe	Primary EE	Russia*	144,586,191	2.3
Eastern Europe	Primary EE	Slovakia*	5,416,406	0.1
Eastern Europe	Primary EE	Slovenia*	2,011,604	0.0
Eastern Europe	Rest of EE	Albania	3,526,642	0.1
Eastern Europe	Rest of EE	Belarus	10,322,151	0.2
Eastern Europe	Rest of EE	Bosnia and Herzegovina	3,989,018	0.1
Eastern Europe	Rest of EE	Estonia	1,350,722	0.0
Eastern Europe	Rest of EE	Latvia	2,322,943	0.0
Eastern Europe	Rest of EE	Lithuania	3,620,094	0.1
Eastern Europe	Rest of EE	Macedonia	2,035,145	0.0
Eastern Europe	Rest of EE	Moldova	4,439,502	0.1
Eastern Europe	Rest of EE	Ukraine	48,055,439	0.8
Eastern Europe	Rest of EE	Yugoslavia (Serbia/Montenegro)	10,823,280	0.2
Latin America	Primary LA	Argentina*	38,740,807	0.6
Latin America	Primary LA	Brazil	182,032,604	2.9

TABLE 2

Worldwide Geographic Regions, Country Affiliations, and Population, 2003

Region	Subregion	Country	Population	Share of Worldwide (%)
Latin America	Primary LA	Chile*	15,662,645	0.2
Latin America	Primary LA	Colombia*	41,662,073	0.7
Latin America	Primary LA	Mexico*	103,718,062	1.6
Latin America	Primary LA	Venezuela*	24,654,694	0.4
Latin America	Rest of LA	Antigua and Barbuda	67,897	0.0
Latin America	Rest of LA	Bahamas	297,477	0.0
Latin America	Rest of LA	Barbados	277,264	0.0
Latin America	Rest of LA	Belize	266,440	0.0
Latin America	Rest of LA	Bolivia	8,586,443	0.1
Latin America	Rest of LA	Costa Rica	3,896,092	0.1
Latin America	Rest of LA	Cuba	11,268,976	0.2
Latin America	Rest of LA	Dominica	69,655	0.0
Latin America	Rest of LA	Dominican Republic	8,715,602	0.1
Latin America	Rest of LA	Ecuador	13,074,082	0.2
Latin America	Rest of LA	El Salvador	6,470,379	0.1
Latin America	Rest of LA	Grenada	89,258	0.0
Latin America	Rest of LA	Guatemala	13,909,384	0.2
Latin America	Rest of LA	Guyana	761,181	0.0
Latin America	Rest of LA	Haiti	7,771,090	0.1
Latin America	Rest of LA	Honduras	6,669,789	0.1
Latin America	Rest of LA	Jamaica	2,695,867	0.0
Latin America	Rest of LA	Nicaragua	5,253,603	0.1
Latin America	Rest of LA	Panama	2,960,784	0.0
Latin America	Rest of LA	Paraguay	6,036,900	0.1
Latin America	Rest of LA	Peru	27,158,869	0.4
Latin America	Rest of LA	Puerto Rico	3,878,679	0.1
Latin America	Rest of LA	St. Kitts and Nevis	41,459	0.0
Latin America	Rest of LA	St. Vincent and Grenadines	118,356	0.0

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TABLE 2

Worldwide Geographic Regions, Country Affiliations, and Population, 2003

Region	Subregion	Country	Population	Share of Worldwide (%)
Latin America	Rest of LA	St. Lucia	164,163	0.0
Latin America	Rest of LA	Suriname	435,449	0.0
Latin America	Rest of LA	Trinidad and Tobago	1,104,209	0.0
Latin America	Rest of LA	Uruguay	3,381,606	0.1
Middle East/Africa	Primary MEA	Egypt, Arab Rep.*	74,718,797	1.2
Middle East/Africa	Primary MEA	Israel*	6,116,533	0.1
Middle East/Africa	Primary MEA	Saudi Arabia*	25,156,630	0.4
Middle East/Africa	Primary MEA	South Africa*	44,481,901	0.7
Middle East/Africa	Primary MEA	Turkey*	68,109,469	1.1
Middle East/Africa	Rest of MEA	Afghanistan	2,484,818	0.0
Middle East/Africa	Rest of MEA	Algeria	27,060,359	0.4
Middle East/Africa	Rest of MEA	Angola	69,150	0.0
Middle East/Africa	Rest of MEA	Armenia	10,766,471	0.2
Middle East/Africa	Rest of MEA	Azerbaijan	3,001,712	0.0
Middle East/Africa	Rest of MEA	Bahrain	7,830,764	0.1
Middle East/Africa	Rest of MEA	Benin	667,238	0.0
Middle East/Africa	Rest of MEA	Botswana	7,041,490	0.1
Middle East/Africa	Rest of MEA	Burkina Faso	1,636,213	0.0
Middle East/Africa	Rest of MEA	Burundi	13,228,460	0.2
Middle East/Africa	Rest of MEA	Cameroon	6,096,156	0.1
Middle East/Africa	Rest of MEA	Cape Verde	15,746,179	0.3
Middle East/Africa	Rest of MEA	Central African Republic	412,137	0.0
Middle East/Africa	Rest of MEA	Chad	3,683,538	0.1
Middle East/Africa	Rest of MEA	Comoros	9,253,493	0.1
Middle East/Africa	Rest of MEA	Congo, Dem Rep. (Kinshasa)	632,948	0.0
Middle East/Africa	Rest of MEA	Congo, Rep. (Brazzaville)	2,954,258	0.0
Middle East/Africa	Rest of MEA	Cote d'Ivoire	56,625,039	0.9
Middle East/Africa	Rest of MEA	Cyprus	16,595,981	0.3

TABLE 2

Worldwide Geographic Regions, Country Affiliations, and Population, 2003

Region	Subregion	Country	Population	Share of Worldwide (%)
Middle East/Africa	Rest of MEA	Djibouti	771,657	0.0
Middle East/Africa	Rest of MEA	Equatorial Guinea	457,130	0.0
Middle East/Africa	Rest of MEA	Eritrea	510,473	0.0
Middle East/Africa	Rest of MEA	Ethiopia	4,362,254	0.1
Middle East/Africa	Rest of MEA	Gabon	69,628,079	1.1
Middle East/Africa	Rest of MEA	Gambia	1,321,560	0.0
Middle East/Africa	Rest of MEA	Ghana	4,710,921	0.1
Middle East/Africa	Rest of MEA	Guinea	20,467,747	0.3
Middle East/Africa	Rest of MEA	Guinea-Bissau	9,030,220	0.1
Middle East/Africa	Rest of MEA	Iran	1,360,827	0.0
Middle East/Africa	Rest of MEA	Iraq	67,147,618	1.1
Middle East/Africa	Rest of MEA	Jordan	24,683,313	0.4
Middle East/Africa	Rest of MEA	Kazakhstan	5,460,265	0.1
Middle East/Africa	Rest of MEA	Kenya	15,107,581	0.2
Middle East/Africa	Rest of MEA	Kuwait	32,168,495	0.5
Middle East/Africa	Rest of MEA	Kyrgyzstan	2,183,161	0.0
Middle East/Africa	Rest of MEA	Lebanon	5,019,150	0.1
Middle East/Africa	Rest of MEA	Lesotho	3,727,703	0.1
Middle East/Africa	Rest of MEA	Liberia	1,861,959	0.0
Middle East/Africa	Rest of MEA	Libya	3,317,176	0.1
Middle East/Africa	Rest of MEA	Madagascar	5,499,074	0.1
Middle East/Africa	Rest of MEA	Malawi	16,979,744	0.3
Middle East/Africa	Rest of MEA	Mali	11,651,239	0.2
Middle East/Africa	Rest of MEA	Malta	11,626,219	0.2
Middle East/Africa	Rest of MEA	Mauritania	395,178	0.0
Middle East/Africa	Rest of MEA	Mauritius	2,912,584	0.0
Middle East/Africa	Rest of MEA	Morocco	1,210,447	0.0
Middle East/Africa	Rest of MEA	Mozambique	31,689,265	0.5

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TABLE 2

Worldwide Geographic Regions, Country Affiliations, and Population, 2003

Region	Subregion	Country	Population	Share of Worldwide (%)
Middle East/Africa	Rest of MEA	Namibia	18,800,908	0.3
Middle East/Africa	Rest of MEA	Nauru	1,993,482	0.0
Middle East/Africa	Rest of MEA	Niger	12,570	0.0
Middle East/Africa	Rest of MEA	Nigeria	11,058,590	0.2
Middle East/Africa	Rest of MEA	Oman	122,796,038	2.0
Middle East/Africa	Rest of MEA	Palau	2,807,125	0.0
Middle East/Africa	Rest of MEA	Palestine	19,717	0.0
Middle East/Africa	Rest of MEA	Qatar	2,626,453	0.0
Middle East/Africa	Rest of MEA	Reunion	817,052	0.0
Middle East/Africa	Rest of MEA	Rwanda	755,171	0.0
Middle East/Africa	Rest of MEA	Sao Tome and Principe	28,119	0.0
Middle East/Africa	Rest of MEA	Senegal	175,883	0.0
Middle East/Africa	Rest of MEA	Seychelles	10,580,307	0.2
Middle East/Africa	Rest of MEA	Sierra Leone	80,469	0.0
Middle East/Africa	Rest of MEA	Somalia	5,732,681	0.1
Middle East/Africa	Rest of MEA	Sudan	8,025,190	0.1
Middle East/Africa	Rest of MEA	Swaziland	38,114,160	0.6
Middle East/Africa	Rest of MEA	Syria	1,161,219	0.0
Middle East/Africa	Rest of MEA	Tajikistan	17,585,540	0.3
Middle East/Africa	Rest of MEA	Tanzania	6,863,752	0.1
Middle East/Africa	Rest of MEA	Togo	35,354,845	0.6
Middle East/Africa	Rest of MEA	Tonga	5,429,299	0.1
Middle East/Africa	Rest of MEA	Tunisia	108,141	0.0
Middle East/Africa	Rest of MEA	Turkmenistan	9,873,149	0.2
Middle East/Africa	Rest of MEA	Tuvalu	4,775,544	0.1
Middle East/Africa	Rest of MEA	Uganda	11,305	0.0
Middle East/Africa	Rest of MEA	United Arab Emirates (UAE)	25,556,448	0.4
Middle East/Africa	Rest of MEA	Uzbekistan	25,981,647	0.4

TABLE 2

Worldwide Geographic Regions, Country Affiliations, and Population, 2003

Region	Subregion	Country	Population	Share of Worldwide (%)
Middle East/Africa	Rest of MEA	Yemen, Rep.	19,349,881	0.3
Middle East/Africa	Rest of MEA	Zambia	10,799,220	0.2
Middle East/Africa	Rest of MEA	Zimbabwe	12,576,742	0.2
North America	Primary NA	Canada*	32,207,113	0.5
North America	Primary NA	United States*	290,342,554	4.6
Western Europe	Primary WE	Austria*	8,162,656	0.1
Western Europe	Primary WE	Belgium*	10,330,824	0.2
Western Europe	Primary WE	Denmark*	5,394,138	0.1
Western Europe	Primary WE	Finland*	5,204,405	0.1
Western Europe	Primary WE	France*	60,180,529	1.0
Western Europe	Primary WE	Germany*	82,398,326	1.3
Western Europe	Primary WE	Greece*	10,625,945	0.2
Western Europe	Primary WE	Ireland*	3,924,023	0.1
Western Europe	Primary WE	Italy*	57,998,353	0.9
Western Europe	Primary WE	Netherlands*	16,223,248	0.3
Western Europe	Primary WE	Norway*	4,555,400	0.1
Western Europe	Primary WE	Portugal*	10,479,955	0.2
Western Europe	Primary WE	Spain*	40,217,413	0.6
Western Europe	Primary WE	Sweden*	8,970,306	0.1
Western Europe	Primary WE	Switzerland*	7,408,319	0.1
Western Europe	Primary WE	United Kingdom*	60,094,648	1.0
Western Europe	Rest of WE	Iceland	291,064	0.0
Western Europe	Rest of WE	Luxembourg	456,764	0.0
Western Europe	Rest of WE	Monaco	32,130	0.0
Worldwide	Total	Total	6,294,171,826	100.0

* Baseline countries for which IDC maintains comprehensive IT market data

Source: IDC and World Bank, 2004

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Variables in the IDC Professional Developer Model

The objective of the IDC Professional Developer Model is to estimate the number of professional developers by country over time. Because the number of professional developers as a proportion to total population and as a ratio of tertiary education rates and software spending is reasonably well understood in a small number of countries that account for a majority of professional developers, it is possible to estimate or predict the number of developers for other countries.

The approach for predicting the number of developers relies on the technique of regression analysis. Regression analysis relies on strong relationships between one or more independent variables, whose values are generally well known, to predict the values of the dependent variables, whose values are not otherwise well known. Therefore, the basic relationship between the dependent and independent variables can be expressed as follows: dependent variable = f (independent variables). IDC believes a strong relationship exists between the number of professional developers, higher education of the population expressed as a percentage, and economic investment (in this case, software spending in the specific geographic areas).

This belief stems from empirical research that has consistently demonstrated that the complexities of software development require a level of conceptual understanding that typically is not addressed in academia until after high school. At the same time, a country's ability to support professional developers is related to its economic circumstances. Software development and use is accompanied by significant prerequisites that include the existence of an advanced infrastructure (electrical and communications) and industrial base so that there is the capability to produce and/or consume software. Consequently, we can now define a relationship between the dependent and the independent variables as follows: professional developers = f (population, tertiary education, software spending).

The objective of this model engine is to define a number of relationships that will enable the construction of the unknown dependent variable (i.e., professional developers) from known independent variables (i.e., population, tertiary education, and software spending).

The independent variables that serve as input to the worldwide developers model can be outputs of other models, and therefore they may have some modeling inherent in their construction. The independent variables used by the worldwide developer model include worldwide population, tertiary education, and software spending. In some cases, these variables have been extended using statistical methods (e.g., interpolation, compound annual growth, or recent annual growth) to provide the coverage needed to complete the array of data required by the engine. All extensions to independent variables are included in the narrative for each variable. These variables are defined in greater detail in the sections that follow. The problem of building models on models is especially inherent in our estimates of the number of professional developers in the "rest of" countries listed in Table 2. Both their software spending and tertiary education rates have been modeled. However, these modeled numbers only represent about 15% of the total forecast by the IDC professional developer model engine.

Independent Variables

Country Population

Historical country-level population (CLP) data was obtained from the World Bank for the modeled countries for 2001–2002. IDC builds off this actual country-level data and worldwide United Nations population forecasts for certain regions to forecast country populations.

Tertiary School Education

This variable reflects the percentage of a country's population that is enrolled in some form of tertiary education or that has previously received some amount of tertiary education. A high school diploma or equivalent is a prerequisite for pursuing tertiary education. This variable was selected because tertiary education is highly correlated with the number of professional developers where census data is available.

For those countries for which there was no tertiary education rate information available, an alternative methodology was used. First, countries were segmented by each of the six geographic regions used in IDC research (refer back to Table 1). Within each region, weighting tertiary education rates by population created a composite view of the known tertiary education. This weighted value was then used as a proxy for the missing tertiary education values by geography.

The data used in this version of the model from UNESCO is for 2003 (and earlier years), and IDC forecasts are provided for 2004–2008.

Software Spending

This data can be found in IDC's *Worldwide Black Book,* which reports and forecasts software spending in detail for the baseline countries referred to in the aggregate (refer back to Table 1) and the regions of which they are a part. The difference between the known software spending for each region in total and the known software spending for baseline countries in those regions is used to estimate software spending for the remaining countries within a region. The approach to estimating software spending for the remaining countries is simplistic, based on population. Although this does not account for the differences in software assimilation that exist between countries, over 95% of software spending is accounted for in the baseline countries specified, as are over 85% of developers. Therefore, the potential margin for error in the worldwide value is very small because most of the data is already accounted for.

Dependent Variables

A dependent variable is a variable whose values are predicted by other variables. Therefore, an understanding of how these dependent variables are derived is a prerequisite for gaining the greatest utility from the worldwide developer model and at the same time appreciating its limitations. Although the dependent variables will be defined in this section, the reader may want to refer to the following section on model construction to better understand how the dependent and independent variables are related in the Professional Developer Model engine. Although the only formal dependent variable in this model is the population to yield a developer (PYD), this

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variable is based on the equation defined in the developer-to-education relationship and a transformation of PYD to create the number of professional developers (NPD).

Population-to-Developer Ratio

This variable is simply the ratio of the population of a country to the number of professional developers in the country. This ratio

- Enables us to compare developers across countries on a normalized basis
- Establishes a relationship between population and developers that is used in deriving the number of professional developers in many countries

Table 3 displays a description of the PYD-dependent variable.

TABLE 3

Dependent Variable: Population to Yield a Developer

Category	Description
Definition	The population of a country required to yield one professional developer
Data type	Integer
Periodicity	Annual time series
Unit of measure	Country
Availability	Available for selected countries for which the population and number of professional developers are known; calculated for countries after the derivation of the developer-to-education relationship and the number of professional developers are known
Derivation	Population to yield a developer (PYD) = population/number of professional developers

Source: IDC, 2002

Developer-to-Education Relationship

The developer-to-education relationship is the heart and soul of the IDC Professional Developer Model. The construction of this equation is described in detail in the next section, titled Model Construction. Table 4 displays a description of the developer-to-education relationship–dependent variable.

Number of Professional Developers

The number of professional developers is derived through a reverse engineering process wherein the CLP is divided by the PYD. Table 5 displays a description of the NPD-dependent variable.

TABLE 4

Dependent Variable: Developer-to-Education Relationship

Category	Description
Definition	Rectangular hyperbola (1/N)-shaped nonlinear regression equation that describes the relationship between the population necessary to account for one developer in the country and the tertiary education of the country
Data type	Formula
Periodicity	1999
Unit of measure	Equation
Availability	Available for selected countries for which the population, number of professional developers, and tertiary education are known; calculated for all countries for which the population and tertiary education are known but the number of developers is not.
Derivation	PYD = -107.90 + 180.18 * (1/(TE - 0.0531))

Note: The number of professional developers (NPD) is derived through a reverse engineering process wherein the CLP is divided by the PYD. Table 5 displays a description of the NPD-dependent variable.

Source: IDC, 2002

TABLE 5

Dependent Variable: Number of Professional Developers

Category	Description
Definition	Worldwide number of professional developers
Data type	Integer (developers)
Periodicity	Annual time series
Unit of measure	Country
Availability	1998–2007
Derivation	Number of professional developers (NPD) = CLP/PYD

Source: IDC, 2002

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Model Construction

The steps for calculating dependent variables are as follows:

1. Collect professional developer benchmark data (from country census data) where possible. The availability of this data generally lags by two or more years, and availability is better for developed countries than for developing countries. IDC selected 1998 as the base year for the analysis due to the significantly higher availability of data compared with any other year.

2. Create the PYD for the known professional developer benchmark data for 1998 (calculations not shown).

3. Build the formulaic relationship (equation) between known PYD and tertiary education based on the known data points. This process was based on nonlinear regression analysis.

4. Use the equation from step 3, known tertiary education data, and population data to reverse engineer the number of professional developers in all countries for 1998.

5. Use the software spending growth (SSG) rate and tertiary education data by country from 1999 through 2008 (calculations not shown) to forecast the number of professional developers from 2001 through 2008.

The Conceptual Model

As far as IDC is able to determine, the number of professional developers is not a data point that is available for the majority of countries. Although most countries collect some data about their labor force, they rarely collect this data to the level of detail outlined above in the Definition section. Thus, we asked a rhetorical question: Is there data that is commonly available for most countries that is also a good predictor through an equation that would enable us to model the number of professional developers?

We believe that the answer to this question is yes, and the results are contained in this study. Some census data regarding the number of professional developers is available as a result of country census efforts, which means we have one of the two primary ingredients necessary to build our model. The other primary ingredient is one or more dependent variables that are widely available across countries and well correlated with (and therefore a good predictor of) the question we have asked. We believe that tertiary education, which is the percentage of population that is enrolled in or has completed some post–high-school education, is a suitable driver (or independent variable) for our model. The majority of professional developers are college graduates. Therefore, countries with high levels of tertiary education will have proportionally more professional developers. At the same time, countries with low levels of tertiary education will have proportionally fewer professional developers.

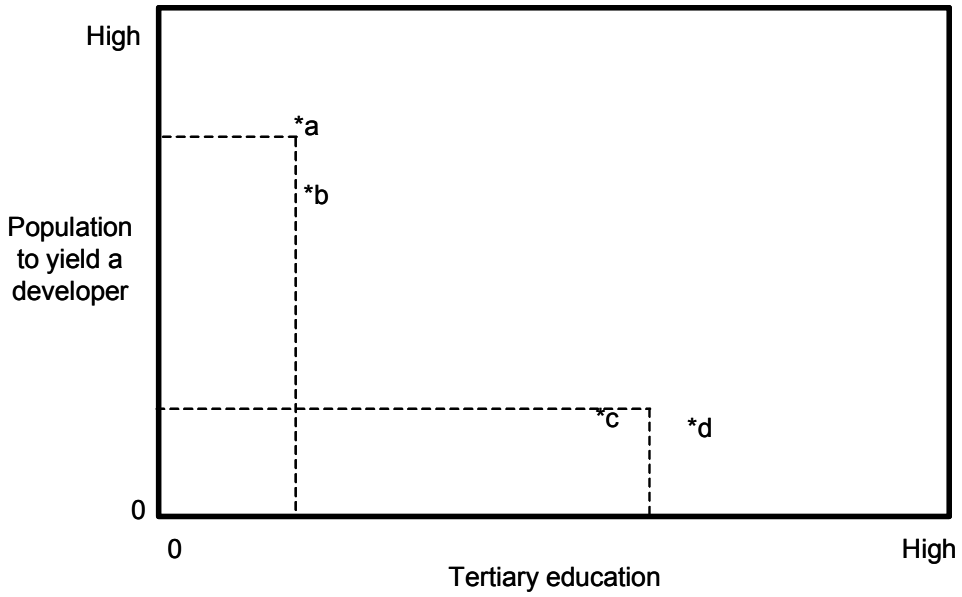
However, we cannot model the number of professional developers based solely on education because of the highly variable populations in the world's countries. Therefore, our model also needs to take into account the population of the country.

The complexity comes in hypothesizing the shape of the relationship between tertiary education and the population of a country necessary to yield a professional developer. The first step in building the model is to plot the known data points. Figure 1 provides an abstract view of what we observed when we plotted the known data points (a, b, c, and d).

As shown in Figure 1, the known data points do not suggest that the optimal least-squares solution would be a straight line. It appears that the optimal solution would be some type of curve that effectively runs asymptotically to both the x-axis and y-axis. In fact, this type of curve fits our expectations because, at near zero levels of tertiary education, we expect a very large, almost infinite PYD, as shown by the dashed lines that approximate the coordinates of data points a and b. The inverse also fits our mental model in that at very high levels of tertiary education, the PYD would be very small and shown by the dashed lines, which approximate the coordinates of data points c and d.

FIGURE 1

Modeled Relationship Between Professional Developers and Tertiary Education



Source: IDC, 2002

We also anticipate that given the position of the data points and the rectangular asymptotic behavior of the relationship, the least-square curve that might best fit these known data points would be a member of the family of curves: Y = 1/X.

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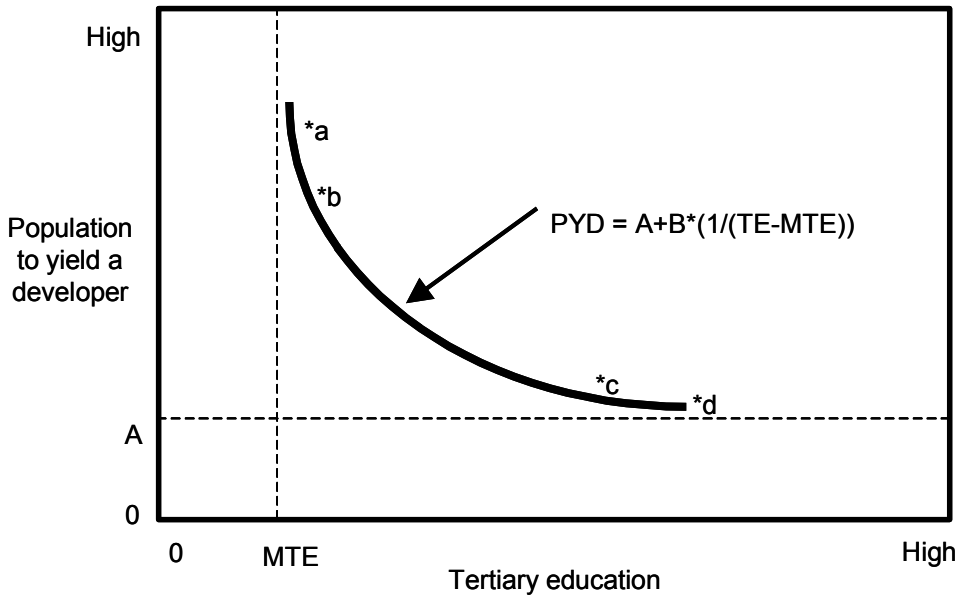
When we superimpose our variables on top of this equation, we have PYD = 1/TE, where PYD is the population to yield one professional developer, and TE is the tertiary education of the country.

We also know that there are practical limits to how this curve can be drawn. We do not believe that there are any countries where no citizens have participated in tertiary education. Similarly, the population needed to yield one professional developer cannot approach one because it is not feasible for every citizen of a country to be a professional developer. Therefore, it is appropriate for us to extend our equation, PYD = 1/TE, so that it contains both a Y intercept (A) and an X intercept (MTE) as well as a multiplier (B) to control the shape of the curve. With the addition of these three modifiers, the equation of our curve now becomes PYD = A + B *(1/(TE - MTE)).

An abstract representation of this equation is shown in Figure 2. This equation is a simple nonlinear regression equation where PYD is the dependent variable. TE is the only independent variable in the equation. MTE is our estimate of the minimum tertiary education for any country. Existing benchmark data will enable us to derive values for A and B by regressing TE on PYD.

FIGURE 2

Modeled Relationship Between Professional Developers and Tertiary Education with Fitted Curve



Source: IDC, 2002

The Physical Model

Until now, we have been describing the process used to build the Professional Developer Model in abstract terms. Let us now transform the conceptual model into a real physical model. The first step in this process is to introduce the benchmark data that will be used in the regression analysis.

As we said, MTE is our estimate of the minimum tertiary education of a country. Because we know historical tertiary education levels in many of the countries, it is not hard to create a reasonable estimate for MTE. The purpose of MTE (as well as A) is to enable us to more accurately position our regression equation because A provides control over the Y intercept (vertical positioning), and MTE provides control over the X intercept (lateral positioning). We have set MTE to 5.31%, which we believe is very close to the minimum TE observed for any country where professional developers are resident.

The key to the 2002–2004 IDC Professional Developer Model is the available census data about the number of professional developers in a country. Because this data is not easy to come by for most developed countries and is virtually nonexistent for undeveloped countries, every incremental data point is a precious commodity. The regression model is built on seven annual data points. The seven data points include census data for the United States in 1998, Canada in 1997, France in 1998, the United Kingdom in 1998, Italy in 1998, India in 1997, and China in 1998. Given the increased frequency of benchmark data in 1998, we used 1998 as the baseline year to which we applied results of our regression analysis. Using this data in our equation resulted in the following values for A and B:

- A = -107.90
- B = 180.18

Although A is a negative number, it does not pose problems because B is a large number, and A and B work together to derive a value for PYD. Because we have elected to set MTE to 5.31% and because we have current and forecast tertiary education values for all countries in 1998, we now have the ability to derive the dependent variable PYD. Thus, our regression equation is now complete as follows: $PYD_{1998} = -107.90 + 180.18 *(1/(TE_{1998} - 0.0531))$.

Once PYD is known, we can leverage the known population for each country to reverse engineer the NPD, as follows: $NPD_{1998} = CLP_{1998}/PYD_{1998}$. Now that the model is built, we can evaluate how it measures up to the known data points. Table 6 displays a comparison of the actual data points and their predicted values. It is evident from the data in Table 6 that the model does a reasonably good job of predicting developers in countries at the extremes of tertiary education. The error margin for Italy is the highest at -15.6%, and the U.S. error margin is lowest at 0.3%. The overall error margin for the data is -1.2%.

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TABLE 6

Comparison of Actual and Predicted Data for the Number of Professional Developers

Country	Actual Data Point	Predicted Data Point	Difference (%)
Canada (1997)	267,000	288,731	8.1
United States (1998)	2,012,550	2,017,923	0.3
France (1998)	250,000	238,580	-4.6
United Kingdom (1998)	239,500	214,028	-10.6
Italy (1998)	200,000	168,770	-15.6
India (1997)	108,258	111,046	2.6
China (1998)	90,000	89,080	-1.0
Total	3,167,308	3,128,158	-1.2

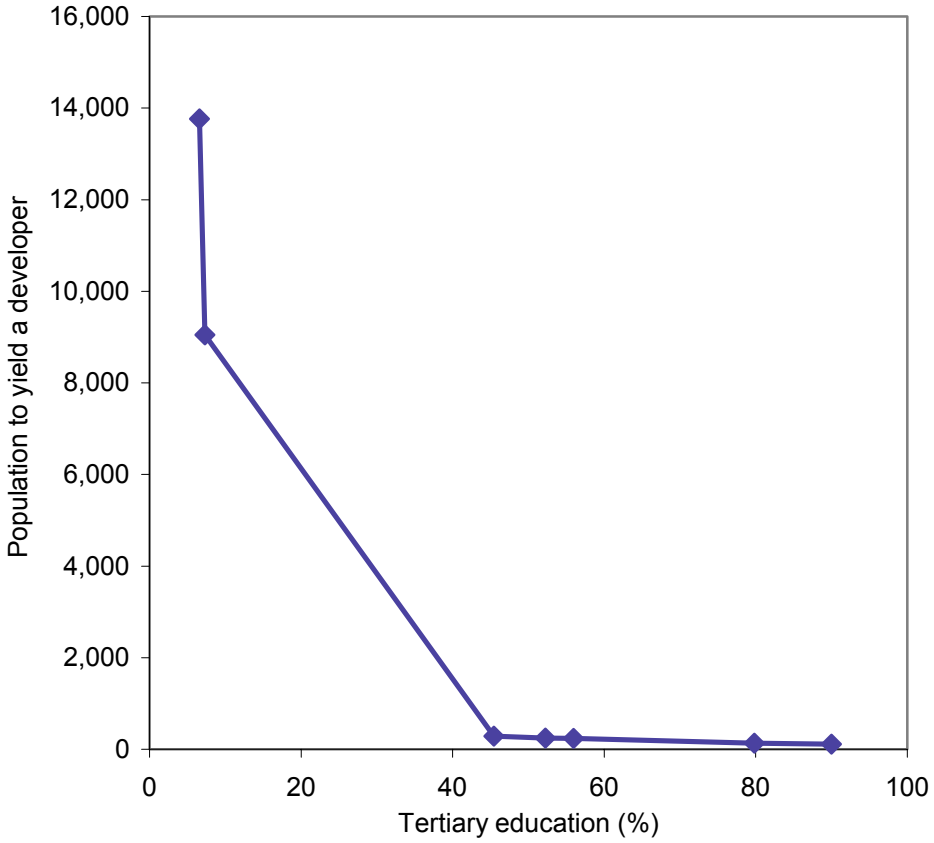
Source: IDC, 2002

Figure 3 displays the actual benchmark data of PYD and TE that was used to build the regression equation. A crude curve made up of line segments has been superimposed over these data points.

Figure 4 uses the same axis and scale as Figure 3, but in this case displays the modeled relationship between PYD and TE. Figure 4 also shows the pronounced knee and inflection point at about 10% TE and 4,000 PYD.

FIGURE 3

Known Professional Developers/Education Relationship



Source: IDC, 2002

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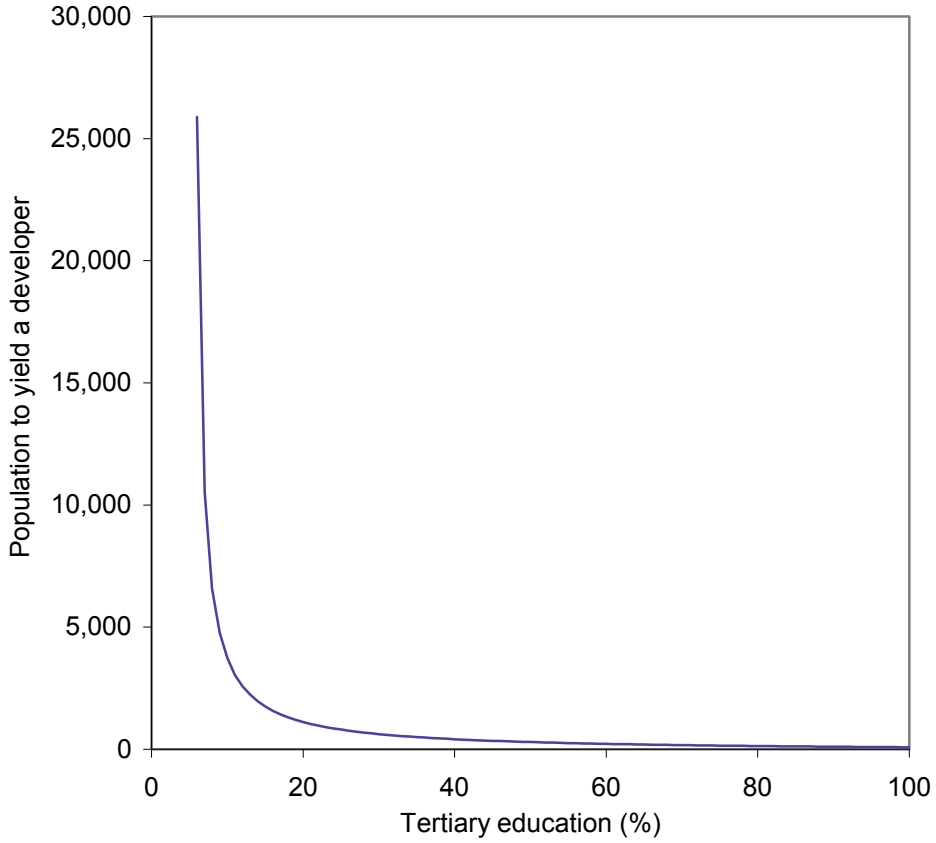
FIGURE 4

Modeled Professional Developers/Education Relationship



Source: IDC, 2002

Forecasting from the Model

As we have said, the modeling effort is based primarily around 1998 because this was the year for which the most government census data concerning professional developers was available. Therefore, a forecasting approach must be employed to define the number of developers for 2001–2008. We elected to use a combination of software spending growth and tertiary education as drivers for this forecast. The forecast equation was defined as follows: $NPD_{N = 1999–2007} = NPD_{N - 1} *(1 + (SSG_N * (1.3 - TE_N)))$.

The effect of this equation is initially to forecast the number of developers based on the growth of software spending that is occurring in the country. The utility of using software spending growth as a primary driver is its ability to reflect in a very granular sense the health of the software industry by country.

However, the propensity of a country to increase its number of developers is also inversely related to its level of tertiary education:

- In developed countries such as the United States or Canada, where tertiary education levels are high, the number of professional developers is also already high relative to the population; growth rates will be low.

- In developing countries such as India and China, tertiary education is low and the number of professional developers is low relative to the population; the increase in both factors will drive higher growth.

- Therefore, we have added the effect of tertiary education into the equation in the following way: $(C - TE_N)$.

In this equation, C is a constant (set to 1.3). The effect of C - TE is to increase the growth of professional developers for countries with low tertiary education levels and decrease the growth of professional developers for countries with high tertiary education levels.

The value of C was selected to minimize the error rate of the forecast through data points for which additional benchmark data was available. The United States and India are two notable cases. Census data for the United States points to 2,555,600 professional developers in 2000. Our forecast for the United States in 2000 was 2,335,962, a -8.6% difference. Likewise, recent data from NASSCOM (a software industry association in India), which was adapted by IDC, suggests that there were 294,274 professional developers in India in 2001. Our forecast for India in 2001 was 303,683, a 3.2% difference.

Note that although the model engine estimates data back to 1998, only the years 2001–2008 are reported in this document per IDC convention.

Summary of Model Assumptions and/or Constraints

Because the IDC Professional Developer Model is just that, a model, it is based on a number of assumptions. These assumptions are itemized as follows:

- Tertiary education, population, and software spending growth are relatively good predictors of the number of professional developers.

- Population forecasts for the baseline countries do not materially change from the current forecasts by the United Nations or World Bank.

- Population in the remaining countries will grow over the forecast period at approximately the same rate as they grew between 1995 and 1999.

- Software spending by major country as described by IDC's *Worldwide Black Book* does not materially change from the current forecasts (November 2004). This is an especially important assumption to watch over time because in fact these numbers do change both because of different time periods and because of the changing feelings of the IT professionals that feed the forecasts.

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- Known tertiary education values for countries within a region can be leveraged to build a composite view for those countries where the tertiary education rate is not known. This is also an important variable to watch; in particular, the estimates of historical penetration rates changed for many countries and in both directions in the most recent UNESCO report prior to the development of the 2004 tables.

- The relationship between tertiary education and the population necessary to yield one professional developer is nonlinear and belongs to the family of curves 1/N.

SITUATION OVERVIEW

Worldwide Professional Developers by Geographic Region

The growth in the number of professional developers worldwide was affected dramatically by the economic downturn in 2001/2002. After increasing 12.0% between 1999 and 2000, the growth rate slowed to 3.2% between 2000 and 2001 (calculations not shown). However, as illustrated in Table 7, the rate rebounded to 6% in 2002 over 2001 (a possible leading indicator of the software spending rebound in 2003) and 7.4% in 2003 over 2002. The total absolute number of developers grew to just over 9.3 million developers worldwide in 2003. Although the world population grows at just over 1%, the number of developers is growing 9.8% compounded annually over the forecast period, and the number of developers will increase from 0.14% of the world's population in 2002 to 0.22% in 2008. In Table 8, more regionally specific growth-rate detail is provided on year-year, region-region growth rates, with each region broken down into two subregions that represent the baseline countries in that region and the rest of the countries that are considered part of that region. As noted in the Methodology section of this document, primary countries are those for which IDC collects the majority of data. In 2003, 85.5% of the developers in the world were in the baseline countries in 2001 (refer back to Table 1).

TABLE 7

Worldwide Population and Professional Developer Estimates, 2001–2008

	2001	2002	2003	2004	2005	2006	2007	2008	2003–2008 CAGR (%)
Population (000)	6,148,527	6,221,760	6,294,753	6,368,037	6,441,693	6,515,614	6,589,793	6,664,003	1.1
Developers (000)	8,201	8,697	9,344	10,147	11,137	12,243	13,499	14,890	9.8
Growth (%)	NA	6.0	7.4	8.6	9.8	9.9	10.3	10.3	
Share of population (%)	0.1	0.1	0.1	0.2	0.2	0.2	0.2	0.2	

Source: IDC Worldwide Professional Developer Model, 2004

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TABLE 8

Worldwide Growth in Professional Developers by Geographic Region, 2001–2008 (%)

	2001	2002	2003	2004	2005	2006	2007	2008
Asia/Pacific								
Primary AP	NA	13.6	9.7	12.0	14.7	15.4	15.8	15.9
Rest of AP	NA	9.2	4.7	9.2	12.0	17.2	21.4	22.1
Subtotal	NA	13.3	9.4	11.9	14.6	15.5	16.1	16.2
Eastern Europe								
Primary EE	NA	25.1	23.9	12.6	11.1	10.2	10.2	9.9
Rest of EE	NA	3.5	3.3	10.8	10.3	8.4	8.9	8.7
Subtotal	NA	20.3	19.9	12.3	10.9	9.9	10.0	9.7
Latin America								
Primary LA	NA	-5.6	7.4	10.0	11.5	10.2	8.4	7.8
Rest of LA	NA	4.9	5.3	9.2	10.9	10.2	8.9	8.6
Subtotal	NA	-1.4	6.5	9.6	11.2	10.2	8.6	8.2
Middle East/Africa								
Primary MEA	NA	5.1	23.5	14.9	14.3	12.5	10.9	9.1
Rest of MEA	NA	7.3	8.8	19.5	21.3	19.2	19.7	19.2
Subtotal	NA	6.8	11.6	18.5	19.9	17.9	18.0	17.4
North America								
Primary NA	NA	-0.6	3.3	3.2	3.4	3.7	3.7	3.3
Subtotal	NA	-0.6	3.3	3.2	3.4	3.7	3.7	3.3
Western Europe								
Primary WE	NA	4.6	2.8	4.5	5.1	5.2	5.0	4.7
Subtotal	NA	4.6	2.8	4.5	5.1	5.2	5.0	4.7
Worldwide								
Baseline countries	NA	6.0	7.5	7.4	8.4	8.7	8.9	8.8
Remaining countries	NA	6.5	7.0	15.5	17.3	16.2	16.9	16.8
Total	NA	6.0	7.4	8.6	9.8	9.9	10.3	10.3

Source: IDC Worldwide Professional Developer Model, 2004

The rates of growth vary substantially between different geographic regions. In Eastern Europe, for example, the growth rate was estimated as 12.3% between 2003 and 2004. Although this was a decrease from the growth rate in Eastern Europe between 2001 and 2002 and between 2002 and 2003 (20.3% and 19.9%, respectively — refer back to Table 8), it was still quite healthy compared with the growth rate in other regions, particularly the 0.6% deflation in North America between 2001 and 2002. Growth in the number of developers did increase in 2003 and 2004 and is forecast to continue to increase modestly. In addition to the boom in Eastern Europe and the maturity in North America, Asia/Pacific continues to grow at a healthy pace, and the Middle East and Africa will eventually see growth rates better than Eastern Europe's, albeit from a very low base.

FUTURE OUTLOOK

As mentioned above, the North American economic downturn of 2000–2002 put significant pressure on the growth in the number of developers worldwide. However, IDC believes the growth rate was actually quite positive in 2002 relative to economic activity, continued to accelerate worldwide in 2003 and 2004, and will get stronger throughout the forecast period, especially in key countries in Asia/Pacific and Eastern Europe (see Table 9). By 2008, IDC forecasts that there will be almost 15 million developers worldwide. (Note that this is heavily influenced by very positive "new" forecasts of population growth and software spending in the out-years of the forecast; we say "new" because this is the first year that 2008 estimates are included. In addition, IDC forecasts for software spending for 2006–2007 are more optimistic than the last time the model engine was generated.) The estimates of absolute numbers translate to a compound annual growth rate (CAGR) of 9.8% for the 2004–2008 forecast period, compared with a CAGR of 8.6% in the 2003 IDC Professional Developer Model (for the 2003–2007 period).

As noted, these increases are totally because of higher expectations for software spending and new historical data for population and tertiary education penetration rates. Our basic assumptions concerning drivers and inhibitors have not changed for this version of the model engine.

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TABLE 9

Worldwide Professional Developers by Country and Region, 2001–2008 (000)

	Subregion	Country	2001	2002	2003	2004	2005	2006	2007	2008	2003–2008 CAGR (%)	2008 Share of Region (%)	2008 Share of World (%)
By country													
Asia/Pacific	Primary AP	Australia	176.5	183.5	188.6	190.7	200.3	210.5	220.8	231.2	4.2	5.3	1.6
Asia/Pacific	Primary AP	China	211.4	264.5	281.5	362.6	459.3	575.4	717.7	878.9	25.6	20.2	5.9
Asia/Pacific	Primary AP	Hong Kong	16.0	14.2	14.0	14.6	15.6	16.8	18.3	20.1	7.5	0.5	0.1
Asia/Pacific	Primary AP	India	303.4	432.5	523.2	639.9	804.2	1,007.5	1,255.9	1,563.0	24.5	35.9	10.5
Asia/Pacific	Primary AP	Indonesia	25.2	42.6	40.1	44.2	50.0	56.5	63.9	71.6	12.3	1.6	0.5
Asia/Pacific	Primary AP	Japan	460.0	457.7	510.6	527.6	544.6	562.0	578.9	595.8	3.1	13.7	4.0
Asia/Pacific	Primary AP	Korea (South)	145.1	163.6	156.3	161.6	168.9	176.8	185.1	194.6	4.5	4.5	1.3
Asia/Pacific	Primary AP	Malaysia	14.5	15.2	14.9	15.9	16.9	18.1	19.5	20.9	7.0	0.5	0.1
Asia/Pacific	Primary AP	New Zealand	19.0	17.1	19.4	19.2	19.9	20.9	22.0	23.0	3.4	0.5	0.2
Asia/Pacific	Primary AP	Philippines	28.6	25.0	21.6	23.4	25.5	28.0	30.8	34.0	9.5	0.8	0.2
Asia/Pacific	Primary AP	Singapore	14.0	12.0	12.2	12.7	13.2	13.9	14.7	15.5	4.8	0.4	0.1
Asia/Pacific	Primary AP	Taiwan	93.8	101.2	103.7	108.5	115.8	124.4	133.5	142.5	6.6	3.3	1.0
Asia/Pacific	Primary AP	Thailand	126.8	129.4	146.4	158.3	180.4	205.7	231.1	255.4	11.8	5.9	1.7
Asia/Pacific	Primary AP	Vietnam	17.1	17.5	25.7	27.3	31.3	35.7	42.1	49.2	13.8	1.1	0.3

TABLE 9

Worldwide Professional Developers by Country and Region, 2001–2008 (000)

	Subregion	Country	2001	2002	2003	2004	2005	2006	2007	2008	2003–2008 CAGR (%)	2008 Share of Region (%)	2008 Share of World (%)
Asia/Pacific	Rest of AP	Bangladesh	3.2	3.5	3.7	4.1	4.6	5.4	6.6	8.1	17.1	0.2	0.1
Asia/Pacific	Rest of AP	Bhutan	0.0	0.1	0.1	0.1	0.1	0.1	0.1	0.1	18.3	0.0	0.0
Asia/Pacific	Rest of AP	Brunei	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	17.0	0.0	0.0
Asia/Pacific	Rest of AP	Cambodia	1.1	1.2	1.3	1.4	1.6	1.9	2.4	3.0	18.1	0.1	0.0
Asia/Pacific	Rest of AP	Fiji	0.4	0.4	0.4	0.4	0.5	0.6	0.7	0.8	15.6	0.0	0.0
Asia/Pacific	Rest of AP	Kiribati	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1	16.0	0.0	0.0
Asia/Pacific	Rest of AP	Korea, Dem. Rep. (North)	76.5	83.4	87.3	95.2	106.4	124.5	150.8	183.6	16.0	4.2	1.2
Asia/Pacific	Rest of AP	Laos	0.2	0.2	0.2	0.2	0.3	0.3	0.4	0.5	17.8	0.0	0.0
Asia/Pacific	Rest of AP	Maldives	0.1	0.1	0.1	0.1	0.1	0.1	0.2	0.2	16.0	0.0	0.0
Asia/Pacific	Rest of AP	Marshall Islands	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	16.0	0.0	0.0
Asia/Pacific	Rest of AP	Micronesia	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1	16.0	0.0	0.0
Asia/Pacific	Rest of AP	Mongolia	2.1	2.3	2.3	2.5	2.8	3.3	3.9	4.6	14.6	0.1	0.0
Asia/Pacific	Rest of AP	Myanmar	6.2	6.8	7.1	7.7	8.7	10.1	12.3	14.9	16.0	0.3	0.1
Asia/Pacific	Rest of AP	Nepal	1.2	1.3	1.4	1.5	1.7	2.0	2.5	3.0	17.4	0.1	0.0
Asia/Pacific	Rest of AP	Pakistan	0.9	1.0	1.1	1.2	1.4	1.6	2.0	2.5	17.7	0.1	0.0

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TABLE 9

Worldwide Professional Developers by Country and Region, 2001–2008 (000)

	Subregion	Country	2001	2002	2003	2004	2005	2006	2007	2008	2003–2008 CAGR (%)	2008 Share of Region (%)	2008 Share of World (%)
Asia/Pacific	Rest of AP	Papua New Guinea	0.2	0.2	0.2	0.2	0.2	0.3	0.4	0.5	17.7	0.0	0.0
Asia/Pacific	Rest of AP	Samoa	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	17.5	0.0	0.0
Asia/Pacific	Rest of AP	Solomon Islands	0.1	0.1	0.2	0.2	0.2	0.2	0.3	0.3	16.0	0.0	0.0
Asia/Pacific	Rest of AP	Sri Lanka	13.7	15.1	15.8	17.3	19.5	23.1	28.4	35.1	17.3	0.8	0.2
Asia/Pacific	Rest of AP	Vanuatu	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	16.0	0.0	0.0
Asia/Pacific	Total	Total	1,757.4	1,991.8	2,179.7	2,438.7	2,794.1	3,226.0	3,745.4	4,353.4	14.8	100.0	29.2
Eastern Europe	Primary EE	Bulgaria	34.8	40.7	46.9	52.9	59.7	67.0	74.0	81.5	11.7	4.5	0.5
Eastern Europe	Primary EE	Croatia	9.6	10.8	11.9	13.4	15.2	17.1	18.9	20.9	12.0	1.2	0.1
Eastern Europe	Primary EE	Czech Republic	17.7	20.2	23.1	26.2	29.7	33.5	37.2	41.2	12.3	2.3	0.3
Eastern Europe	Primary EE	Hungary	19.9	22.3	24.8	27.3	30.0	32.7	35.9	39.2	9.6	2.2	0.3
Eastern Europe	Primary EE	Poland	65.7	74.2	81.4	89.6	98.9	109.2	120.2	131.0	10.0	7.3	0.9
Eastern Europe	Primary EE	Romania	77.6	109.9	125.9	142.2	160.5	180.4	202.5	228.4	12.6	12.7	1.5
Eastern Europe	Primary EE	Russia	347.7	440.9	578.9	653.8	722.9	790.6	867.1	947.4	10.4	52.6	6.4
Eastern Europe	Primary EE	Slovakia	8.8	9.8	11.0	12.2	13.7	15.3	17.0	18.9	11.4	1.0	0.1
Eastern Europe	Primary EE	Slovenia	5.6	6.1	6.6	7.1	7.6	8.1	8.7	9.3	7.0	0.5	0.1

TABLE 9

Worldwide Professional Developers by Country and Region, 2001–2008 (000)

	Subregion	Country	2001	2002	2003	2004	2005	2006	2007	2008	2003–2008 CAGR (%)	2008 Share of Region (%)	2008 Share of World (%)
Eastern Europe	Rest of EE	Albania	2.0	2.1	2.2	2.6	3.0	3.4	3.9	4.4	14.8	0.2	0.0
Eastern Europe	Rest of EE	Belarus	40.7	41.9	43.1	47.0	51.1	54.7	58.7	63.0	7.9	3.5	0.4
Eastern Europe	Rest of EE	Bosnia and Herzegovina	10.4	10.8	11.2	12.5	13.9	15.2	16.6	18.2	10.2	1.0	0.1
Eastern Europe	Rest of EE	Estonia	5.7	5.8	6.0	6.5	7.1	7.6	8.1	8.7	7.7	0.5	0.1
Eastern Europe	Rest of EE	Latvia	6.8	7.0	7.3	8.1	9.0	9.8	10.8	11.8	10.1	0.7	0.1
Eastern Europe	Rest of EE	Lithuania	9.6	10.0	10.3	11.6	12.9	14.1	15.5	17.0	10.5	0.9	0.1
Eastern Europe	Rest of EE	Macedonia	2.9	3.1	3.2	3.7	4.2	4.7	5.3	5.9	12.9	0.3	0.0
Eastern Europe	Rest of EE	Moldova	9.7	10.1	10.5	11.9	13.4	14.7	16.3	18.1	11.5	1.0	0.1
Eastern Europe	Rest of EE	Ukraine	63.4	65.4	67.4	73.9	80.8	86.8	93.7	101.0	8.4	5.6	0.7
Eastern Europe	Rest of EE	Yugoslavia (Serbia/ Montenegro)	18.0	18.9	19.7	22.6	25.7	28.5	31.9	35.5	12.5	2.0	0.2
Eastern Europe	Total	Total	756.5	910.1	1,091.4	1,225.3	1,359.4	1,493.5	1,642.2	1,801.4	10.5	100.0	12.1
Latin America	Primary LA	Argentina	66.6	36.7	45.4	56.4	66.3	75.8	84.2	92.1	15.2	9.6	0.6
Latin America	Primary LA	Brazil	82.1	79.6	74.8	80.6	87.2	94.2	101.7	109.7	8.0	11.4	0.7
Latin America	Primary LA	Chile	26.0	26.8	32.2	34.2	36.4	38.5	40.4	42.2	5.5	4.4	0.3

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TABLE 9

Worldwide Professional Developers by Country and Region, 2001–2008 (000)

	Subregion	Country	2001	2002	2003	2004	2005	2006	2007	2008	2003–2008 CAGR (%)	2008 Share of Region (%)	2008 Share of World (%)
Latin America	Primary LA	Colombia	51.7	53.8	67.3	74.0	82.9	91.0	98.4	105.9	9.5	11.0	0.7
Latin America	Primary LA	Mexico	90.4	109.2	115.6	124.7	133.8	143.2	153.0	163.5	7.2	17.0	1.1
Latin America	Primary LA	Venezuela	33.6	24.7	19.9	20.8	28.7	37.0	42.0	47.1	18.8	4.9	0.3
Latin America	Rest of LA	Antigua and Barbuda	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	11.5	0.0	0.0
Latin America	Rest of LA	Bahamas	0.4	0.4	0.4	0.5	0.5	0.6	0.7	0.7	11.2	0.1	0.0
Latin America	Rest of LA	Barbados	0.5	0.6	0.6	0.7	0.7	0.8	0.9	1.0	11.1	0.1	0.0
Latin America	Rest of LA	Belize	0.2	0.3	0.3	0.3	0.3	0.4	0.4	0.5	12.1	0.1	0.0
Latin America	Rest of LA	Bolivia	10.5	11.1	11.8	13.0	14.7	16.5	18.2	20.1	11.2	2.1	0.1
Latin America	Rest of LA	Costa Rica	16.1	16.8	17.6	19.1	21.0	22.9	24.7	26.6	8.5	2.8	0.2
Latin America	Rest of LA	Cuba	5.6	6.0	6.4	7.1	8.0	9.0	10.0	11.1	11.7	1.2	0.1
Latin America	Rest of LA	Dominica	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	11.5	0.0	0.0
Latin America	Rest of LA	Dominican Republic	10.4	11.0	11.7	12.9	14.6	16.4	18.1	19.9	11.3	2.1	0.1
Latin America	Rest of LA	Ecuador	48.6	50.8	53.3	57.8	63.7	69.8	75.6	81.6	8.9	8.5	0.5
Latin America	Rest of LA	El Salvador	5.2	5.5	5.9	6.5	7.4	8.3	9.1	10.1	11.5	1.1	0.1

TABLE 9

Worldwide Professional Developers by Country and Region, 2001–2008 (000)

	Subregion	Country	2001	2002	2003	2004	2005	2006	2007	2008	2003–2008 CAGR (%)	2008 Share of Region (%)	2008 Share of World (%)
Latin America	Rest of LA	Grenada	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.2	11.5	0.0	0.0
Latin America	Rest of LA	Guatemala	2.5	2.6	2.8	3.1	3.5	4.0	4.4	4.9	11.8	0.5	0.0
Latin America	Rest of LA	Guyana	0.3	0.3	0.3	0.4	0.4	0.5	0.5	0.6	11.7	0.1	0.0
Latin America	Rest of LA	Haiti	1.1	1.1	1.2	1.3	1.5	1.7	1.9	2.1	12.1	0.2	0.0
Latin America	Rest of LA	Honduras	2.3	2.4	2.6	2.9	3.3	3.7	4.1	4.5	11.7	0.5	0.0
Latin America	Rest of LA	Jamaica	0.6	0.6	0.6	0.7	0.8	0.9	1.0	1.1	11.8	0.1	0.0
Latin America	Rest of LA	Nicaragua	2.0	2.1	2.2	2.5	2.8	3.2	3.5	3.9	11.7	0.4	0.0
Latin America	Rest of LA	Panama	11.5	12.0	12.6	13.5	14.7	15.8	16.9	17.9	7.4	1.9	0.1
Latin America	Rest of LA	Paraguay	1.7	1.8	1.9	2.2	2.5	2.8	3.1	3.4	11.8	0.4	0.0
Latin America	Rest of LA	Peru	91.0	95.2	100.1	108.8	120.2	131.9	143.1	154.9	9.1	16.1	1.0
Latin America	Rest of LA	Puerto Rico	11.8	12.4	13.2	14.5	16.3	18.1	19.9	21.8	10.6	2.3	0.1
Latin America	Rest of LA	St. Kitts and Nevis	0.0	0.0	0.0	0.1	0.1	0.1	0.1	0.1	11.5	0.0	0.0
Latin America	Rest of LA	St. Vincent and Grenadines	0.1	0.1	0.1	0.1	0.2	0.2	0.2	0.2	11.5	0.0	0.0
Latin America	Rest of LA	St. Lucia	0.1	0.1	0.2	0.2	0.2	0.2	0.2	0.3	11.5	0.0	0.0

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TABLE 9

Worldwide Professional Developers by Country and Region, 2001–2008 (000)

	Subregion	Country	2001	2002	2003	2004	2005	2006	2007	2008	2003–2008 CAGR (%)	2008 Share of Region (%)	2008 Share of World (%)
Latin America	Rest of LA	Suriname	0.3	0.3	0.3	0.4	0.4	0.5	0.5	0.6	11.6	0.1	0.0
Latin America	Rest of LA	Trinidad and Tobago	0.2	0.2	0.2	0.3	0.3	0.3	0.4	0.4	11.8	0.0	0.0
Latin America	Rest of LA	Uruguay	6.5	6.9	7.3	8.1	9.1	10.2	11.2	12.3	11.0	1.3	0.1
Latin America	Total	Total	580.0	571.8	609.2	667.8	742.9	818.6	888.8	961.4	9.6	100.0	6.5
Middle East/Africa	Primary MEA	Egypt, Arab Rep.	34.1	27.1	28.3	31.5	35.4	41.1	45.7	50.0	12.1	2.1	0.3
Middle East/Africa	Primary MEA	Israel	22.3	22.2	22.3	22.9	23.8	24.9	25.9	26.9	3.8	1.1	0.2
Middle East/Africa	Primary MEA	Saudi Arabia	23.2	33.8	37.6	42.0	46.6	51.9	57.1	62.8	10.8	2.7	0.4
Middle East/Africa	Primary MEA	South Africa	54.5	50.6	80.8	88.4	98.1	109.0	120.9	128.9	9.8	5.5	0.9
Middle East/Africa	Primary MEA	Turkey	29.4	37.7	42.6	58.3	73.9	85.5	96.5	108.6	20.6	4.6	0.7
Middle East/Africa	Rest of MEA	United Arab Emirates (UAE)	2.4	2.9	3.5	4.2	4.9	5.6	6.4	7.3	15.6	0.3	0.0
Middle East/Africa	Rest of MEA	Afghanistan	7.8	8.4	9.2	11.1	13.7	16.5	19.9	24.0	21.1	1.0	0.2
Middle East/Africa	Rest of MEA	Algeria	49.2	52.8	57.6	69.1	84.1	100.6	120.8	144.5	20.2	6.1	1.0
Middle East/Africa	Rest of MEA	Angola	0.8	0.8	0.9	1.1	1.3	1.6	2.0	2.4	21.2	0.1	0.0
Middle East/Africa	Rest of MEA	Armenia	1.2	1.3	1.5	1.8	2.1	2.6	3.1	3.7	20.4	0.2	0.0

TABLE 9

Worldwide Professional Developers by Country and Region, 2001–2008 (000)

	Subregion	Country	2001	2002	2003	2004	2005	2006	2007	2008	2003–2008 CAGR (%)	2008 Share of Region (%)	2008 Share of World (%)
Middle East/Africa	Rest of MEA	Azerbaijan	14.5	15.6	16.9	20.3	24.6	29.4	35.2	42.0	19.9	1.8	0.3
Middle East/Africa	Rest of MEA	Bahrain	1.2	1.3	1.5	1.7	2.1	2.5	3.0	3.6	19.8	0.2	0.0
Middle East/Africa	Rest of MEA	Benin	2.6	2.8	3.1	3.7	4.6	5.5	6.7	8.0	21.0	0.3	0.1
Middle East/Africa	Rest of MEA	Botswana	0.3	0.3	0.4	0.4	0.5	0.7	0.8	0.9	20.8	0.0	0.0
Middle East/Africa	Rest of MEA	Burkina Faso	6.0	6.4	7.0	8.5	10.4	12.6	15.2	18.4	21.2	0.8	0.1
Middle East/Africa	Rest of MEA	Burundi	1.0	1.1	1.2	1.5	1.8	2.2	2.6	3.1	21.2	0.1	0.0
Middle East/Africa	Rest of MEA	Cameroon	5.0	5.3	5.8	7.0	8.6	10.4	12.6	15.1	20.9	0.6	0.1
Middle East/Africa	Rest of MEA	Cape Verde	3.2	3.4	3.6	4.2	5.0	5.8	6.8	7.9	16.9	0.3	0.1
Middle East/Africa	Rest of MEA	Central African Republic	1.0	1.1	1.2	1.4	1.7	2.1	2.5	3.1	21.1	0.1	0.0
Middle East/Africa	Rest of MEA	Chad	3.1	3.4	3.7	4.4	5.4	6.6	8.0	9.6	21.2	0.4	0.1
Middle East/Africa	Rest of MEA	Comoros	0.2	0.2	0.2	0.3	0.4	0.4	0.5	0.6	21.2	0.0	0.0
Middle East/Africa	Rest of MEA	Congo, Dem Rep. (Kinshasa)	28.0	30.1	32.9	39.5	48.3	58.0	69.8	83.8	20.6	3.6	0.6
Middle East/Africa	Rest of MEA	Congo, Rep. (Brazzaville)	1.3	1.4	1.6	1.9	2.3	2.8	3.4	4.1	21.1	0.2	0.0
Middle East/Africa	Rest of MEA	Cote d'Ivoire	6.7	7.2	7.9	9.5	11.7	14.0	17.0	20.4	20.9	0.9	0.1

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TABLE 9

Worldwide Professional Developers by Country and Region, 2001–2008 (000)

	Subregion	Country	2001	2002	2003	2004	2005	2006	2007	2008	2003–2008 CAGR (%)	2008 Share of Region (%)	2008 Share of World (%)
Middle East/Africa	Rest of MEA	Cyprus	2.3	2.5	2.7	3.2	3.9	4.6	5.5	6.6	19.5	0.3	0.0
Middle East/Africa	Rest of MEA	Djibouti	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	21.2	0.0	0.0
Middle East/Africa	Rest of MEA	Equatorial Guinea	0.2	0.3	0.3	0.3	0.4	0.5	0.6	0.7	21.1	0.0	0.0
Middle East/Africa	Rest of MEA	Eritrea	1.2	1.3	1.5	1.8	2.2	2.6	3.1	3.8	21.2	0.2	0.0
Middle East/Africa	Rest of MEA	Ethiopia	19.1	20.6	22.5	27.2	33.4	40.2	48.7	58.7	21.2	2.5	0.4
Middle East/Africa	Rest of MEA	Gabon	0.7	0.8	0.8	1.0	1.2	1.5	1.8	2.2	20.6	0.1	0.0
Middle East/Africa	Rest of MEA	Gambia	0.6	0.6	0.7	0.8	1.0	1.2	1.4	1.7	21.1	0.1	0.0
Middle East/Africa	Rest of MEA	Ghana	5.8	6.3	6.9	8.3	10.2	12.3	14.8	17.9	21.1	0.8	0.1
Middle East/Africa	Rest of MEA	Guinea	4.0	4.3	4.7	5.7	7.0	8.4	10.2	12.3	21.1	0.5	0.1
Middle East/Africa	Rest of MEA	Guinea-Bissau	16.4	17.4	18.7	21.8	25.8	30.0	35.1	40.8	16.9	1.7	0.3
Middle East/Africa	Rest of MEA	Iran	58.5	62.8	68.3	81.7	99.2	118.3	141.7	169.0	19.9	7.2	1.1
Middle East/Africa	Rest of MEA	Iraq	25.3	27.2	29.6	35.6	43.4	52.0	62.5	74.9	20.4	3.2	0.5
Middle East/Africa	Rest of MEA	Jordan	16.5	17.5	18.7	21.6	25.2	28.9	33.1	37.7	15.1	1.6	0.3
Middle East/Africa	Rest of MEA	Kazakhstan	71.2	76.1	82.5	97.7	117.5	138.9	164.8	194.9	18.8	8.3	1.3
Middle East/Africa	Rest of MEA	Kenya	9.6	10.4	11.4	13.7	16.8	20.3	24.6	29.6	21.1	1.3	0.2

TABLE 9

Worldwide Professional Developers by Country and Region, 2001–2008 (000)

	Subregion	Country	2001	2002	2003	2004	2005	2006	2007	2008	2003–2008 CAGR (%)	2008 Share of Region (%)	2008 Share of World (%)
Middle East/Africa	Rest of MEA	Kuwait	5.3	5.6	6.1	7.3	8.9	10.6	12.7	15.1	19.8	0.6	0.1
Middle East/Africa	Rest of MEA	Kyrgyzstan	6.6	7.1	7.7	9.3	11.3	13.5	16.3	19.5	20.3	0.8	0.1
Middle East/Africa	Rest of MEA	Lebanon	7.8	8.3	9.1	10.8	13.0	15.4	18.3	21.7	19.2	0.9	0.1
Middle East/Africa	Rest of MEA	Lesotho	0.3	0.3	0.3	0.4	0.5	0.6	0.7	0.8	21.0	0.0	0.0
Middle East/Africa	Rest of MEA	Liberia	1.0	1.1	1.2	1.4	1.8	2.1	2.6	3.1	21.0	0.1	0.0
Middle East/Africa	Rest of MEA	Libya	3.2	3.4	3.7	4.4	5.4	6.4	7.6	9.1	19.7	0.4	0.1
Middle East/Africa	Rest of MEA	Madagascar	5.2	5.6	6.1	7.3	9.0	10.8	13.1	15.8	21.1	0.7	0.1
Middle East/Africa	Rest of MEA	Malawi	2.6	2.8	3.1	3.7	4.6	5.5	6.7	8.1	21.2	0.3	0.1
Middle East/Africa	Rest of MEA	Mali	2.7	2.9	3.1	3.8	4.7	5.6	6.8	8.2	21.2	0.3	0.1
Middle East/Africa	Rest of MEA	Malta	1.8	1.9	2.1	2.5	3.0	3.5	4.2	5.0	19.0	0.2	0.0
Middle East/Africa	Rest of MEA	Mauritania	0.9	1.0	1.1	1.3	1.6	1.9	2.3	2.8	20.9	0.1	0.0
Middle East/Africa	Rest of MEA	Mauritius	0.4	0.4	0.4	0.5	0.6	0.7	0.9	1.1	20.7	0.0	0.0
Middle East/Africa	Rest of MEA	Morocco	42.4	45.6	49.7	59.7	72.8	87.2	104.9	125.6	20.4	5.3	0.8
Middle East/Africa	Rest of MEA	Mozambique	4.1	4.4	4.9	5.9	7.2	8.7	10.5	12.7	21.2	0.5	0.1
Middle East/Africa	Rest of MEA	Namibia	2.0	2.1	2.3	2.8	3.4	4.1	4.9	5.9	20.5	0.2	0.0

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TABLE 9

Worldwide Professional Developers by Country and Region, 2001–2008 (000)

	Subregion	Country	2001	2002	2003	2004	2005	2006	2007	2008	2003–2008 CAGR (%)	2008 Share of Region (%)	2008 Share of World (%)
Middle East/Africa	Rest of MEA	Nauru	0.2	0.2	0.3	0.3	0.4	0.4	0.5	0.6	16.9	0.0	0.0
Middle East/Africa	Rest of MEA	Niger	2.0	2.2	2.4	2.9	3.6	4.3	5.2	6.3	21.2	0.3	0.0
Middle East/Africa	Rest of MEA	Nigeria	18.4	19.8	21.6	26.1	32.0	38.5	46.5	56.0	20.9	2.4	0.4
Middle East/Africa	Rest of MEA	Oman	1.8	1.9	2.1	2.5	3.1	3.7	4.5	5.4	20.6	0.2	0.0
Middle East/Africa	Rest of MEA	Palau	0.2	0.2	0.2	0.2	0.3	0.3	0.4	0.4	16.9	0.0	0.0
Middle East/Africa	Rest of MEA	Palestine	22.9	24.4	26.2	30.6	36.1	42.0	49.1	57.2	16.9	2.4	0.4
Middle East/Africa	Rest of MEA	Qatar	2.8	3.1	3.3	3.9	4.8	5.6	6.7	8.0	19.2	0.3	0.1
Middle East/Africa	Rest of MEA	Reunion	7.1	7.6	8.1	9.5	11.2	13.0	15.2	17.7	16.9	0.8	0.1
Middle East/Africa	Rest of MEA	Rwanda	0.9	1.0	1.1	1.4	1.7	2.0	2.4	2.9	21.2	0.1	0.0
Middle East/Africa	Rest of MEA	Sao Tome and Principe	2.3	2.5	2.6	3.1	3.6	4.2	5.0	5.8	16.9	0.2	0.0
Middle East/Africa	Rest of MEA	Senegal	3.7	3.9	4.3	5.2	6.4	7.7	9.3	11.2	21.0	0.5	0.1
Middle East/Africa	Rest of MEA	Seychelles	0.8	0.8	0.9	1.0	1.2	1.4	1.7	1.9	16.9	0.1	0.0
Middle East/Africa	Rest of MEA	Sierra Leone	2.0	2.1	2.3	2.8	3.5	4.2	5.1	6.1	21.1	0.3	0.0
Middle East/Africa	Rest of MEA	Somalia	0.8	0.9	1.0	1.1	1.4	1.7	2.1	2.5	21.1	0.1	0.0
Middle East/Africa	Rest of MEA	Sudan	21.3	22.9	25.1	30.2	37.1	44.6	53.9	64.8	20.9	2.8	0.4

TABLE 9

Worldwide Professional Developers by Country and Region, 2001–2008 (000)

	Subregion	Country	2001	2002	2003	2004	2005	2006	2007	2008	2003–2008 CAGR (%)	2008 Share of Region (%)	2008 Share of World (%)
Middle East/Africa	Rest of MEA	Swaziland	0.3	0.3	0.4	0.4	0.5	0.6	0.8	0.9	20.8	0.0	0.0
Middle East/Africa	Rest of MEA	Syria	34.4	37.0	40.3	48.2	58.6	70.1	84.1	100.5	20.1	4.3	0.7
Middle East/Africa	Rest of MEA	Tajikistan	20.1	21.5	23.4	27.9	33.9	40.3	48.3	57.5	19.7	2.4	0.4
Middle East/Africa	Rest of MEA	Tanzania	7.0	7.5	8.2	9.9	12.2	14.7	17.8	21.5	21.2	0.9	0.1
Middle East/Africa	Rest of MEA	Togo	2.2	2.3	2.5	3.1	3.8	4.5	5.5	6.6	21.0	0.3	0.0
Middle East/Africa	Rest of MEA	Tonga	1.2	1.3	1.4	1.6	1.9	2.2	2.6	3.0	16.9	0.1	0.0
Middle East/Africa	Rest of MEA	Tunisia	14.9	16.0	17.5	20.9	25.5	30.5	36.6	43.7	20.2	1.9	0.3
Middle East/Africa	Rest of MEA	Turkmenistan	10.1	10.9	11.8	14.1	17.1	20.4	24.4	29.1	19.7	1.2	0.2
Middle East/Africa	Rest of MEA	Tuvalu	0.1	0.1	0.1	0.1	0.1	0.2	0.2	0.2	16.9	0.0	0.0
Middle East/Africa	Rest of MEA	Uganda	9.5	10.2	11.2	13.5	16.6	19.9	24.1	29.1	21.1	1.2	0.2
Middle East/Africa	Rest of MEA	Uzbekistan	13.2	14.1	15.2	18.0	21.6	25.5	30.2	35.6	18.5	1.5	0.2
Middle East/Africa	Rest of MEA	Yemen, Rep.	6.3	6.7	7.4	8.9	10.9	13.1	15.8	19.1	20.9	0.8	0.1
Middle East/Africa	Rest of MEA	Zambia	3.3	3.6	3.9	4.7	5.8	7.0	8.4	10.1	21.0	0.4	0.1
Middle East/Africa	Rest of MEA	Zimbabwe	3.7	4.0	4.4	5.3	6.5	7.8	9.4	11.3	20.7	0.5	0.1
Middle East/Africa	Total	Total	850.0	908.2	1,013.8	1,201.3	1,440.0	1,697.2	2,003.5	2,353.1	18.3	100.0	15.8

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TABLE 9

Worldwide Professional Developers by Country and Region, 2001–2008 (000)

	Subregion	Country	2001	2002	2003	2004	2005	2006	2007	2008	2003–2008 CAGR (%)	2008 Share of Region (%)	2008 Share of World (%)
North America	Primary NA	Canada	335.1	330.3	360.8	366.8	374.0	382.6	392.9	404.2	2.3	12.5	2.7
North America	Primary NA	United States	2,315.9	2,304.7	2,361.6	2,441.4	2,530.0	2,629.7	2,729.6	2,821.3	3.6	87.5	18.9
North America	Total	Total	2,651.0	2,635.0	2,722.4	2,808.2	2,903.9	3,012.3	3,122.4	3,225.6	3.4	100.0	21.7
Western Europe	Primary WE	Austria	34.0	35.5	35.1	36.5	38.1	39.9	41.7	43.3	4.3	2.0	0.3
Western Europe	Primary WE	Belgium	53.2	55.5	61.0	63.5	66.5	69.7	73.0	76.1	4.5	3.5	0.5
Western Europe	Primary WE	Denmark	20.6	22.3	24.8	25.9	27.1	28.4	29.6	30.8	4.5	1.4	0.2
Western Europe	Primary WE	Finland	37.4	36.5	40.6	41.9	43.5	45.2	46.8	48.3	3.6	2.2	0.3
Western Europe	Primary WE	France	269.8	282.5	277.6	289.3	303.2	318.2	333.3	348.3	4.6	15.9	2.3
Western Europe	Primary WE	Germany	303.6	315.0	349.0	365.4	385.2	406.8	429.1	451.1	5.3	20.5	3.0
Western Europe	Primary WE	Greece	35.9	37.2	31.3	32.6	34.2	36.0	37.8	39.7	4.9	1.8	0.3
Western Europe	Primary WE	Ireland	13.1	13.7	13.2	14.0	14.9	16.0	17.1	18.1	6.5	0.8	0.1
Western Europe	Primary WE	Italy	199.9	211.3	215.2	225.2	237.3	250.8	264.9	279.3	5.4	12.7	1.9
Western Europe	Primary WE	Netherlands	70.8	74.2	69.1	72.1	75.5	79.1	82.9	86.6	4.6	3.9	0.6
Western Europe	Primary WE	Norway	28.7	27.1	30.3	31.5	33.0	34.5	35.9	37.3	4.2	1.7	0.3
Western Europe	Primary WE	Portugal	29.8	31.3	27.9	29.2	30.9	32.8	34.8	36.7	5.6	1.7	0.2

TABLE 9

Worldwide Professional Developers by Country and Region, 2001–2008 (000)

	Subregion	Country	2001	2002	2003	2004	2005	2006	2007	2008	2003–2008 CAGR (%)	2008 Share of Region (%)	2008 Share of World (%)
Western Europe	Primary WE	Spain	186.1	195.9	200.1	211.3	223.5	236.0	248.5	261.4	5.5	11.9	1.8
Western Europe	Primary WE	Sweden	40.6	40.8	42.7	44.2	46.1	48.0	49.9	51.6	3.9	2.4	0.3
Western Europe	Primary WE	Switzerland	18.5	19.5	22.5	23.5	24.7	26.1	27.5	28.9	5.1	1.3	0.2
Western Europe	Primary WE	United Kingdom	264.0	281.8	287.1	299.1	313.0	328.0	343.2	358.2	4.5	16.3	2.4
Western Europe	Total	Total	1,606.2	1,680.0	1,727.4	1,805.3	1,896.7	1,995.4	2,096.1	2,195.5	4.9	100.0	14.7
By region													
Asia/Pacific	Primary AP	Total	1,651.3	1,875.9	2,058.4	2,306.3	2,645.9	3,052.1	3,534.4	4,095.8	14.8	NA	27.5
Asia/Pacific	Rest of AP	Total	106.1	115.9	121.3	132.4	148.3	173.8	211.0	257.6	16.3	NA	1.7
Eastern Europe	Primary EE	Total	587.3	735.0	910.5	1,024.8	1,138.3	1,254.0	1,381.4	1,517.8	10.8	NA	10.2
Eastern Europe	Rest of EE	Total	169.2	175.1	180.9	200.5	221.1	239.6	260.8	283.6	9.4	NA	1.9

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TABLE 9

Worldwide Professional Developers by Country and Region, 2001–2008 (000)

	Subregion	Country	2001	2002	2003	2004	2005	2006	2007	2008	2003–2008 CAGR (%)	2008 Share of Region (%)	2008 Share of World (%)
Latin America	Primary LA	Total	350.3	330.8	355.3	390.7	435.4	479.7	519.7	560.5	9.5	NA	3.8
Latin America	Rest of LA	Total	229.7	241.1	253.9	277.2	307.5	338.9	369.1	400.9	9.6	NA	2.7
Middle East/Africa	Primary MEA	Total	163.3	171.3	211.6	243.1	277.8	312.3	346.1	377.1	12.2	NA	2.5
Middle East/Africa	Rest of MEA	Total	686.7	736.9	802.2	958.2	1,162.2	1,384.9	1,657.5	1,975.9	19.8	NA	13.3
North America	Primary NA	Total	2,651.0	2,635.0	2,722.4	2,808.2	2,903.9	3,012.3	3,122.4	3,225.6	3.4	NA	21.7
Western Europe	Primary WE	Total	1,606.2	1,680.0	1,727.4	1,805.3	1,896.7	1,995.4	2,096.1	2,195.5	4.9	NA	14.7
Worldwide	Total	Total	8,201.1	8,697.0	9,343.9	10,146.6	11,137.1	12,243.1	13,498.6	14,890.2	9.8	NA	100.0

Source: IDC Worldwide Professional Developer Model, 2004

Outlook by Region

North America

North American year-over-year employment growth rates for professional developers will be low compared with the growth rates in other regions, but the number will grow steadily from a large base. After two successive years of professional-developer employment deflation in North America (2000 and 2001), the number of developers in North America increased slightly in 2003 and 2004. As would be expected for such a large economy, which has historically been the home of most developers, growth rates will remain in low single digits during the forecast period. This is to be expected as the calculation of the number of developers in each country is in part a function of the growth in software spending in that country, which is also lower for North America than other regions in the world. Program development automation techniques, offshoring, and increasing productivity are all likely to put downward pressure on the rate at which the number of developers in North America increases during the forecast period.

Western Europe and Latin America

Growth in Western Europe is expected to remain muted. The number of developers in Western Europe will increase at a CAGR of 4.9% during the forecast period, while Latin America will increase at a CAGR of 9.6%. As established in the 2001 IDC Professional Developer Model and continued into the 2002, 2003, and 2004 methodology, the model is built on the premise that countries and regions that are more developed will generally have less growth in developers than developing countries and regions. This trend is due to the S-shaped curve that typically characterizes technology adoption.

Asia/Pacific

Significant changes in the distribution of developers worldwide during the forecast period will occur in the Asia/Pacific region and Eastern Europe. Asia/Pacific is expected to have a CAGR during the forecast period of 14.8%. As a result, Asia/Pacific will have the largest share of developers by the end of the forecast period, surpassing North America during the forecast period. Given that China and India alone represent more than one-third of the world's population and that many vendors are looking to Asia/Pacific to pick up in terms of IT spending (and low-cost 24 x 7 support for the wired world), this rise to dominance is expected.

Eastern Europe

The Eastern European countries, led by those that recently joined the European Union, will soon find the resources to increase their infrastructure necessary to enable the widespread adoption of technology. Therefore, the region will see a spike in the demand for developers during the midyears of the forecast period. Once this infrastructure is in place, technology adoption and growth in the number of professional developers accelerates rapidly. Then, once technology becomes well established and there is an adequate cadre of professional developers available, growth trails off and the region becomes more like Western Europe in professional-developer-model dynamics.

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Middle East and Africa

The Middle East and Africa have been inhibited in IT aspects by its political instability, but the number of developers has nevertheless managed to continue to increase from a relatively small base. By the end of the forecast period, the model says that the Middle East and Africa will account for over 15% of all programmers worldwide, but this number, translating to an almost 20% CAGR, is somewhat driven by the growth in the overall population rate, which may be overstating the possibilities for this region.

Professional Developer Penetration Ratio by Region

It is important to remember that the IDC Professional Developer Model is just that, a model (refer back to Table 9). Therefore, the risk of potential errors is most pronounced at this most detailed level of the model. As mentioned previously, however, IDC continues to collect data that will improve the model, and we will adjust the methodology as relationships between variables become clearer.

Another way to evaluate the growth in professional developers over time is by using a metric we call the professional developer penetration ratio. This ratio is simply the population of a country or region divided by the number of professional developers for the country or region.

The values in Table 10 show the relationship between the number of developers and the total population by country. Notice how the values in the table decrease over time. This decrease indicates that the number of professional developers for the region is increasing relative to the population.

For developing regions, a rapid decrease in this ratio generally means rapid growth in the number of professional developers. This is almost universally true, in fact, because there are few countries or regions where the population is declining. For North America in 2004, where 1 person out of every 116 people is already a professional developer, the ratio decreases slowly during the forecast period. For the nonbaseline portion of Asia/Pacific region, however, the ratio will decline from 1 out of 4,226 in 2001 to 1 out of 1,956 in 2008 (in the latter example, the absolute numbers differ dramatically from the 2003 model engine results because of different historic data inputs; however, the trend line remains the same).

As in earlier models, Western Europe's ratio is almost double that of North America. Nevertheless, the data shows that the ratio of developers to population in these developed regions is substantially higher than in the rest of the world.

TABLE 10

Worldwide Population per Developer by Geographic Region, 2001–2008

	2001	2002	2003	2004	2005	2006	2007	2008
Asia/Pacific								
Primary AP	1,801.6	1,602.5	1,475.4	1,330.1	1,170.9	1,025.0	893.7	778.7
Rest of AP	4,225.6	3,934.2	3,819.3	3,557.7	3,230.5	2,802.9	2,348.6	1,956.0
Subtotal	1,947.9	1,738.2	1,605.8	1,451.0	1,280.2	1,120.8	975.7	848.4
Eastern Europe								
Primary EE	420.6	335.0	269.5	238.8	214.4	194.1	175.7	159.5
Rest of EE	539.1	518.7	500.3	449.8	406.5	374.0	342.5	314.2
Subtotal	447.1	370.3	307.8	273.3	245.6	222.9	202.2	183.9
Latin America								
Primary LA	1,132.0	1,213.9	1,144.1	1,053.0	955.8	877.5	818.9	767.7
Rest of LA	571.2	553.1	533.4	496.0	453.8	418.0	389.5	363.8
Subtotal	909.9	935.3	889.5	821.8	748.0	687.2	640.6	599.3
Middle East/Africa								
Primary MEA	1,296.2	1,251.8	1,027.4	905.8	802.1	721.7	658.4	610.4
Rest of MEA	1,368.5	1,304.6	1,226.7	1,050.1	885.3	759.3	648.3	555.5
Subtotal	1,354.4	1,294.5	1,184.4	1,020.4	868.9	752.3	650.1	564.5
North America								
Primary NA	119.4	121.3	118.5	115.9	113.1	110.1	107.1	104.6
Subtotal	119.4	121.3	118.5	115.9	113.1	110.1	107.1	104.6
Western Europe								
Primary WE	242.9	232.9	227.0	217.7	207.7	197.8	188.6	180.4
Subtotal	242.9	232.9	227.0	217.7	207.7	197.8	188.6	180.4
Worldwide								
Baseline countries	647.5	616.7	578.9	543.7	506.0	469.6	435.1	403.2
Remaining countries	1,352.4	1294.5	1232.9	1088.1	945.6	829.1	722.5	630.1
Total	749.7	715.4	673.7	627.6	578.4	532.2	488.2	447.5

Source: IDC Worldwide Professional Developer Model, 2004

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Software Spending, Professional Developers, and Productivity

Table 11 contains data on the ratio of worldwide software spending per developer from 2001 through 2008. This data is generated somewhat as a byproduct of the process of building the model because software spending is one of the variables used to forecast the growth rates in the number of developers by country. As it turns out, this data provides some interesting insights into the relationship between technological development and productivity (but on the other hand, these results could be a "circular reference").

Software spending per developer continues to trend down over the forecast period on a worldwide basis (as it trends up in Western Europe and North America, where both IT developers and users will invest more in all aspects of developer automation technology to support fewer increasingly costly professional developers). Further, the data shows that, in developing regions (including Asia/Pacific, Latin America, and the Middle East and Africa), the growth in the number of developers outstrips the growth in software spending. This data seems to indicate that vendors that sell relatively inexpensive developer tools might have an easier time penetrating developing markets compared with those vendors that sell more expensive enterprise software.

TABLE 11

Worldwide Software Spending per Developer by Geographic Region, 2001–2008 ($)

	2001	2002	2003	2004	2005	2006	2007	2008
Asia/Pacific								
Primary AP	13,743.7	12,493.8	12,495.8	11,837.0	11,067.5	10,322.5	9,604.1	8,930.1
Rest of AP	235.7	233.0	231.6	229.1	225.8	221.5	216.5	211.4
Subtotal	12,928.3	11,780.4	11,813.3	11,206.8	10,492.2	9,778.2	9,075.2	8,414.1
Eastern Europe								
Primary EE	3,342.2	3,238.2	3,182.7	3,247.0	3,327.0	3,409.8	3,487.8	3,556.8
Rest of EE	1,773.1	1,798.6	1,822.4	1,896.0	1,967.4	2,027.5	2,091.9	2,156.0
Subtotal	2,991.3	2,961.2	2,957.3	3,026.0	3,105.9	3,188.1	3,266.1	3,336.2
Latin America								
Primary LA	9,449.1	9,473.9	8,942.3	8,839.0	8,703.1	8,617.4	8,582.2	8,563.2
Rest of LA	1,088.3	1,086.8	1,085.2	1,082.9	1,080.3	1,077.9	1,075.9	1,073.9
Subtotal	6,137.7	5,938.0	5,667.6	5,620.1	5,548.0	5,496.1	5,465.1	5,440.1
Middle East/Africa								
Primary MEA	12,151.2	12,030.5	11,965.6	11,499.0	11,150.1	10,931.0	10,778.4	10,635.5
Rest of MEA	584.6	579.0	570.8	554.6	537.7	522.6	507.6	493.2
Subtotal	2,840.1	2,774.9	2,989.2	2,807.4	2,621.3	2,472.4	2,314.6	2,150.2
North America								
Primary NA	35,022.1	34,917.4	35,412.0	36,369.8	37,447.1	38,678.0	39,927.2	41,091.4
Subtotal	35,022.1	34,917.4	35,412.0	36,369.8	37,447.1	38,678.0	39,927.2	41,091.4
Western Europe								
Primary WE	31,189.3	31,626.8	32,258.2	32,680.0	33,166.9	33,681.1	34,196.2	34,676.3
Subtotal	31,189.3	31,626.8	32,258.2	32,680.0	33,166.9	33,681.1	34,196.2	34,676.3
Worldwide								
Baseline countries	24,661.2	23,710.3	23,344.0	23,076.4	22,752.3	22,460.5	22,111.5	21,665.0
Remaining countries	819.8	812.7	803.9	792.6	775.8	755.5	732.9	710.3
Total	21,203.9	20,376.7	20,076.1	19,641.6	19,133.2	18,681.6	18,164.7	17,568.7

Source: IDC Worldwide Professional Developer Model, 2004

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ESSENTIAL GUIDANCE

The data contained in this study can be interpreted in many ways, depending on one's perspective. However, there are two primary conclusions that can be drawn as follows:

- An application development and deployment software vendor can address the large majority of the world's professional developers by targeting the development and deployment needs of the 10 countries with the largest developer populations: the United States, Russia, India, Japan, Canada, Germany, China, the United Kingdom, France, and Italy (with Spain almost moving up to 10th place).

- Total geographic specificity may be a mistake in the "brave new one world" that is emerging interconnected, highly collaborative, and following the sun. The following should be considered:

  - Vendors should provide any geographic specific requirements of the tools to the extent that they end up in the finished application in one of the leading economies.

  - IT demands may emanate from the leading economies mentioned above.

  - Developers could just as easily be residents of Iceland or New Zealand.

Of course, this is not meant to imply that vendors should forsake the North American market. The United States will continue to be the home to the largest number of professional developers in the world for the foreseeable future, with the model estimating 2.8 million in 2008 (18.9% of the total worldwide), compared with 1.6 million for second-place India (10%). In addition, despite the maturity — or perhaps because of the maturity — of the United States' market, demographics dictate a large turnover of the North American professional developer population over the next 20 years as the first of the "professional IT developer class," who are also baby boomers, move into retirement. The turnover in North America in professional developers is not measured in this document, only the net result; that turnover could provide more opportunity than the millions of pure head count coming online in other regions.

Application development and deployment vendors looking for smaller, high-growth markets have a number of opportunities available to them. In particular, many developing countries will need additional training opportunities as well as software and documentation in their native languages. Those vendors that are willing to address these smaller markets can potentially garner a larger share of those markets than would be possible in the largest markets, which will be targeted by the majority of vendors. But it is important to remember that the resulting software still needs to be deployed to support the largest economies.

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- *Worldwide Black Book Query Tool, Version 3, 2004* (IDC #32322, November 2004)
- *Worldwide Software Market Forecast Summary, 2004–2008* (IDC #31785, September 2004)
- *The 2003 IDC Professional Developer Model* (IDC #30534, December 2003)

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